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Uni-Perceiver: Pre-training Unified Architecture for Generic Perception for Zero-shot and Few-shot Tasks

Xizhou Zhu[1*], Jinguo Zhu[2*†], Hao Li[3*†], Xiaoshi Wu[3*†]
Xiaogang Wang[3], Hongsheng Li[3], Xiaohua Wang[2], Jifeng Dai[1✉]
[1]SenseTime Research [2]Xi'an Jiaotong University
[3]CUHK-SenseTime Joint Laboratory, The Chinese University of Hong Kong
{zhuwalter, daijifeng}@sensetime.com
lechatelia@stu.xjtu.edu.cn, {haoli, wuxiaoshi}@link.cuhk.edu.hk
{xgwang, hsli}@ee.cuhk.edu.hk, xhw@mail.xjtu.edu.cn

Abstract

Biological intelligence systems of animals perceive the world by integrating information in different modalities and processing simultaneously for various tasks. In contrast, current machine learning research follows a task-specific paradigm, leading to inefficient collaboration between tasks and high marginal costs of developing perception models for new tasks. In this paper, we present a generic perception architecture named Uni-Perceiver, which processes a variety of modalities and tasks with unified modeling and shared parameters. Specifically, Uni-Perceiver encodes different task inputs and targets from arbitrary modalities into a unified representation space with a modality-agnostic Transformer encoder and lightweight modality-specific tokenizers. Different perception tasks are modeled as the same formulation, that is, finding the maximum likelihood target for each input through the similarity of their representations. The model is pre-trained on several uni-modal and multi-modal tasks, and evaluated on a variety of downstream tasks, including novel tasks that did not appear in the pre-training stage. Results show that our pre-trained model without any tuning can achieve reasonable performance even on novel tasks. The performance can be improved to a level close to state-of-the-art methods by conducting prompt tuning on 1% of downstream task data. Full-data fine-tuning further delivers results on par with or better than state-of-the-art results. Code shall be released.

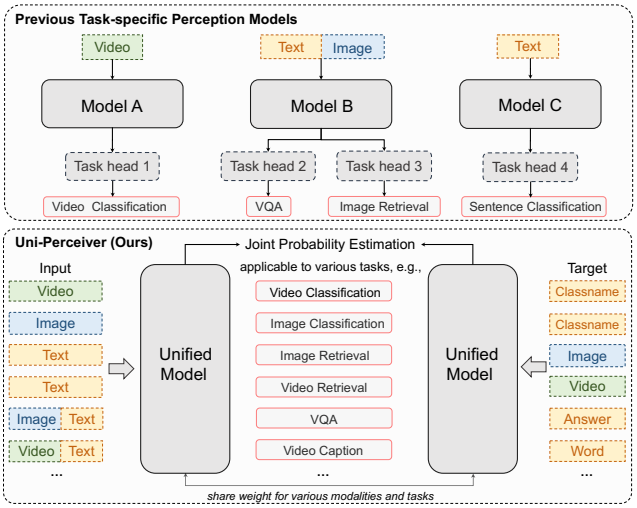


Figure 1. Comparing previous task-specific perception models with our proposed Uni-Perceiver, which processes various modalities and tasks with a single siamese model and shared parameters.

1. Introduction

Biological intelligence systems of animals perceive the world by receiving information in different modalities, integrating with the complex central nervous system, and processing simultaneously for different tasks. However, designing a generic artificial perception model that handles multiple modalities and numerous tasks has always been considered too difficult. To simplify this problem, previous machine learning research has focused on developing specialized models for inputs from certain restricted modality, *e.g.*, Convolutional Neural Networks [39] for visual recognition and Transformers [69] for natural language processing. Recently, Transformers have been proved to

*Equal contribution. †This work is done when Jinguo Zhu, Hao Li, and Xiaoshi Wu are interns at SenseTime Research. ✉Corresponding author.

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Uni-Perceiver：面向零样本和少样本任务的通用感知统一架构预训练

朱熹周[1*]，朱金国[2*†]，李昊[3*†]，吴晓诗[3*†] 王晓刚[3]，李鸿升[3]，王晓华[2]，戴季峰[1✉] [1]商汤科技研究院 [2]西安交通大学 [3]香港中文大学-商汤科技联合实验室，香港中文大学

{zhuwalter, daijifeng}@sensetime.com lechatelia@stu.xjtu.edu.cn, {haoli, wuxiaoshi}@link.cuhk.edu.hk {xgwang, hsli}@ee.cuhk.edu.hk, xhw@mail.xjtu.edu.cn

摘要

动物的生物智能系统通过整合不同模态的信息并同时处理各种任务来感知世界。相比之下，当前的机器学习研究遵循任务特定的范式，导致任务之间的协作效率低下，并且为新任务开发感知模型的边际成本高昂。本文提出了一种名为 Uni-Perceiver 的通用感知架构，它通过统一建模和共享参数来处理多种模态和任务。具体而言，Uni-Perceiver 使用模态无关的 Transformer 编码器和轻量级的模态特定分词器，将来自任意模态的不同任务输入和目标编码到统一的表示空间中。不同的感知任务被建模为相同的形式，即通过输入与目标表示之间的相似性，为每个输入寻找最大似然目标。该模型在若干单模态和多模态任务上进行预训练，并在多种下游任务上进行评估，包括预训练阶段未出现过的新任务。结果表明，我们的预训练模型在无需任何调优的情况下，即使在新任务上也能取得合理的性能。通过在 1% 的下游任务数据上进行提示调优，性能可提升到接近最先进方法的水平。使用全量数据进行微调则进一步取得了与最先进结果相当或更优的结果。代码将会发布。

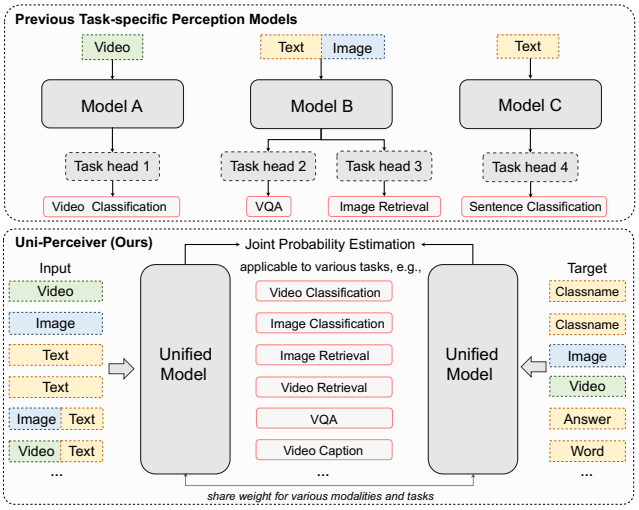


图 1. 将以往特定任务的感知模型与我们提出的 Uni-Perceiver 进行比较，后者使用单一的孪生模型和共享参数处理各种模态和任务。

1. 引言

动物的生物智能系统通过接收不同模态的信息、与复杂的中枢神经系统进行整合，并同时针对不同任务进行处理来感知世界。然而，设计一个能够处理多种模态和众多任务的通用人工感知模型一直被认为过于困难。为了简化这一问题，以往的机器学习研究主要集中于开发针对某些受限模态输入的专用模型，例如用于视觉识别的卷积神经网络 [39] 和用于自然语言处理的 Transformer [69]。最近，Transformer 已被证明能够

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have competitive performance in more scenarios such as image [9, 17, 45, 66, 68, 71, 73, 76] and video [4, 5, 74] recognition, which triggers a new paradigm of designing unified architectures for different modalities. Following this paradigm, recent works [1, 22, 28, 56] adopt Transformers as the backbone for multi-modal applications such as visual-linguistic recognition. They convert the inputs from different modalities into unified input token sequences with modality-specific tokenizers. Models are pre-trained with large-scale multi-modal datasets, and then adapted to downstream tasks with fine-tuning.

Despite the ability of processing multi-modal information with unified architectures, current methods still require specific design and training for different tasks. This limitation is caused by two reasons. First, the input of a particular model is the combination of specific modalities required by its target task. Second, previous works require prediction heads specifically designed and trained for the target tasks.

We argue that this task-specific paradigm conflicts with the objective of designing generic perceptual models. Specifically, during pre-training, the specialised designs for different tasks hinder the collaboration between tasks, which may hurt the representational capacity. Meanwhile, when a pre-trained model is applied to a new task, the input format and the prediction head need to be re-designed and fine-tuned on sufficient downstream data. Considerable effort in collecting and annotating data is required. Also, all parameters need to be copied and maintained for each downstream task, which becomes inefficient and inconvenient as the number of tasks and the model size grow. On the other hand, when fine-tuning is performed with insufficient training data, it may forget the pre-trained knowledge that is beneficial to the downstream task, thereby hurting generalization performance [13]. All of these issues increase the marginal cost of developing perception models for new tasks and limit the capability to meet the rapidly growing demands of diverse scenarios, indicating task-specific paradigm is not suitable for generic perceptual modeling.

Our core idea is to replace task-specific designs by encoding different task inputs and targets from arbitrary modalities into a unified representation space, and model the joint probability of inputs and targets through the similarity of their representations. This design eliminates the gap between the formulations of different perception tasks, and therefore encourages the collaboration between different modalities and tasks in representation learning. Moreover, by aligning the formulations of pre-training and downstream tasks, the knowledge can be better transferred when applying the pre-trained model to the target tasks. The model can even conduct zero-shot inference on novel tasks that do not appear in the pre-training stage.

In this paper, we propose a unified architecture named Uni-Perceiver, which processes various modalities and tasks with a single siamese model and shared parameters. Specifically, the task inputs and targets from arbitrary combinations of modalities are first converted into unified token sequences with lightweight modality-specific tokenizers. The sequences are then encoded by a modality-agnostic Transformer encoder into a unified representation space. Different perception tasks are modeled as the same formulation, finding the maximum likelihood target for each input through the similarity of their representations, so as to facilitate the generic perceptual modeling.

Uni-Perceiver is pre-trained on various uni-modal tasks such as image / video classification and language modeling, and multi-modal tasks such as image-text retrieval and language modeling with image clues. When applied to downstream tasks, thanks to the generic modeling of perception tasks, the pre-trained model shows the ability of zero-shot inference on novel tasks that did not appear in the pre-training stage. Moreover, the performance can be further boosted with additional task-specific data. For the few-shot scenario, we adapt the model to downstream tasks with prompt tuning [41], where only a small amount of additional parameters are optimized for specific tasks. The performance of our model can be further improved with full-model fine-tuning on sufficient downstream training data.

We pre-train our model on several uni-modal and multi-modal tasks, and evaluate its performance on a variety of downstream tasks, including novel tasks that did not appear in the pre-training stage. Results show that our pre-trained model without any tuning can achieve reasonable performance even on novel tasks. Its performance can be boosted to a level close to state-of-the-art methods by conducting prompt tuning with 1% of the downstream task data. When fine-tuning the pre-trained model with 100% of the target data, our model achieves result on par with or better than state-of-the-art methods on almost all the tasks, which demonstrates the strong representation ability.

2. Related Works

Architecture. For visual recognition, Convolutional Neural Networks (CNN) [39] used to be the main architecture paradigm. Motivated by the success of Transformers in natural language processing [7, 16, 31, 34, 44, 69], attempts have been made to apply Transformers to image and video modalities. For image recognition, vision Transformers [9, 17, 45, 66, 68, 71, 73, 76] replace CNNs by an image patch tokenizer and a transformer encoder, which have been proved to obtain competitive performance as CNNs. [4, 5, 74] make attempts to apply Transformers on video recognition in a convolution-free fashion. For visual-linguistic recognition, recent works [14, 36, 37, 46, 53, 62, 65, 78] also adopt Transformers as the backbone, while they usually take regional features as inputs, which are typically extracted by off-the-shelf object detectors (*e.g.*, Faster R-

在图像 [9, 17, 45, 66, 68, 71, 73, 76] 和视频 [4, 5, 74] 识别等更多场景中具有竞争力的性能，这促成了一种为不同模态设计统一架构的新范式。遵循这一范式，近期工作 [1, 22, 28, 56] 采用 Transformer 作为视觉-语言识别等多模态应用的骨干网络。它们使用特定于模态的分词器，将来自不同模态的输入转换为统一的输入 token 序列。模型先在大规模多模态数据集上进行预训练，然后通过微调适配下游任务。

尽管统一架构具备处理多模态信息的能力，但当前方法仍然需要针对不同任务进行特定的设计和训练。这一局限性由两个原因造成。首先，某一特定模型的输入是其目标任务所需的特定模态的组合。其次，先前的工作需要为目标任务专门设计并训练预测头。

我们认为，这种特定任务范式与设计通用感知模型的目标相冲突。具体而言，在预训练过程中，针对不同任务的专门化设计会阻碍任务之间的协作，从而可能损害表征能力。同时，当将预训练模型应用于新任务时，需要重新设计输入格式和预测头，并在充足的下游数据上进行微调。这需要在数据收集和标注方面投入大量工作。此外，对于每个下游任务，都需要复制并维护所有参数；随着任务数量和模型规模的增长，这会变得低效且不便。另一方面，当在训练数据不足的情况下进行微调时，模型可能会遗忘对下游任务有益的预训练知识，从而损害泛化性能 [13]。所有这些问题都会增加为新任务开发感知模型的边际成本，并限制其满足快速增长的多样化场景需求的能力，这表明特定任务范式并不适合通用感知建模。

我们的核心思想是用一种统一的表示空间来编码来自任意模态的不同任务输入和目标，从而取代针对特定任务的设计，并通过它们表示之间的相似性来建模输入与目标的联合概率。这种设计消除了不同感知任务表述之间的差距，因此促进了不同模态和任务在表示学习中的协同。此外，通过对齐预训练和下游任务的表述，在将预训练模型应用于目标任务时，知识能够得到更好的迁移。该模型甚至可以对预训练阶段未出现的新任务进行零样本推理。

在本文中，我们提出了一种名为 Uni-Perceiver 的统一架构，用于处理各种模态和使用一个共享参数的单一孪生模型处理任务。具体而言，来自任意模态组合的任务输入和目标首先通过轻量级的模态特定分词器转换为统一的标记序列。随后，这些序列由模态无关的 Transformer 编码器编码到统一的表示空间中。不同的感知任务被建模为相同的形式，即通过其表示之间的相似性，为每个输入寻找最大似然目标，从而促进通用感知建模。

Uni-Perceiver 在各种单模态任务上进行预训练，例如图像/视频分类和语言建模；也在多模态任务上进行预训练，例如图像-文本检索以及带有图像线索的语言建模。当应用于下游任务时，得益于对感知任务的通用建模，该预训练模型展现出对预训练阶段未出现的新任务进行零样本推理的能力。此外，借助额外的任务特定数据，性能还可以进一步提升。对于少样本场景，我们通过提示调优 [41] 将模型适配到下游任务，其中仅优化少量额外参数以适应特定任务。在具备充足下游训练数据的情况下，通过全模型微调，我们模型的性能还可以进一步提高。

我们在多个单模态和多模态任务上对模型进行预训练，并在多种下游任务上评估其性能，包括预训练阶段未出现过的新任务。结果表明，即使不进行任何调优，我们的预训练模型也能在新任务上取得合理的性能。通过使用 1% 的下游任务数据进行提示调优，其性能可提升至接近最先进方法的水平。当使用 100% 的目标数据对预训练模型进行微调时，我们的模型在几乎所有任务上都取得了与最先进方法相当或更优的结果，这证明了其强大的表征能力。

2. 相关工作

架构。对于视觉识别，卷积神经网络（CNN）[39] 曾是主要的架构范式。受 Transformer 在自然语言处理 [7, 16, 31, 34, 44, 69] 中成功的启发，人们尝试将 Transformer 应用于图像和视频模态。对于图像识别，视觉 Transformer [9, 17, 45, 66, 68, 71, 73, 76] 通过图像块分词器和 Transformer 编码器取代 CNN，并已被证明能够获得与 CNN 相媲美的性能。[4, 5, 74] 尝试以无卷积的方式将 Transformer 应用于视频识别。对于视觉-语言识别，近期工作 [14, 36, 37, 46, 53, 62, 65, 78] 也采用 Transformer 作为骨干网络，但它们通常将区域特征作为输入，而这些特征通常由现成的目标检测器（例如 Faster R-）提取。

CNN [57] pre-trained on Visual Genome [30]). [24] attempts to eliminate the need for object detectors by directly extracting features from the raw pixels with CNNs. [1, 22, 28, 56] take a further step by applying Transformers to raw image patches and word tokens. Transformers have enabled a unified architecture paradigm for different modalities, which only need the modality-specific tokenizers to convert inputs from different modalities into unified input token sequences.

Nevertheless, previous architecture requires prediction heads specifically designed and trained for different perception tasks. Instead, we replace the task-specific design by encoding different task inputs and targets into a unified representation space, and model their relationship by representational similarity. This modification enables our model to conduct zero-shot inference even on novel downstream tasks that did not appear in the pre-training stage.

Pre-training. Large-scale pre-training has achieved great success in the field of deep learning, which can alleviate the data-hungry challenge and improve the performance of downstream tasks [72]. For image recognition, pre-training is usually performed on image classification datasets, *e.g.*, ImageNet [15]. Video recognition networks are either pre-trained on image classification or video classification datasets, *e.g.*, Moments in Time [49] and Kinetics [27]. In natural language processing, self-supervised language modeling [7, 16, 31, 34, 44] is adopted for pre-training on large-scale unlabeled corpora [54]. Specifically, GPT [7] performs the auto-regressive pre-training, which optimizes the probability of the next word conditioned on previous words. BERT [16] uses masked language modeling (MLM) and next sentence prediction (NSP) for pre-training. These pre-trained models can serve as robust feature extractors for downstream tasks with small architecture modifications.

Recent years have witnessed interest in large-scale cross-modal pre-training [72]. Compared with uni-modal pre-training, cross-modal pre-training needs to align information from different modalities. Such pre-training is usually performed on image-text pairs collected from Internet [8, 26, 50, 60] and manual annotated visual-linguistic datasets [30, 40, 50]. Moreover, various pre-training objectives are also proposed to utilize these datasets effectively. The most widely used objectives are image-text retrieval [2, 37, 47, 55, 63, 64, 65], masked language modeling with image clues [2, 37, 47, 62, 63, 64, 65], and masked region modeling [14, 47, 62, 63, 65]. Among them, masked region modeling requires regional features extracted by off-the-shelf object detectors. More recently, CLIP [55] has verified the effectiveness of only performing image-text retrieval pre-training on huge webly collected data.

Previous multi-task pre-training requires task-specific heads, which hinders the collaboration among different tasks. Instead, we encode different task inputs and targets into a unified representation space, and model their relationship by a unified representational similarity, which enables the collaboration between different modalities and tasks. Our pre-training tasks include image and video classification, language modeling with and without image clues, and image-text retrieval. We do not use regional features and the corresponding pre-training tasks.

Prompt Tuning. As an alternative solution to fine-tuning, prompt tuning has recently been proposed in the NLP community, which originated from prompting methods [41]. In prompting, specially designed natural language tokens, or namely prompts, are inserted into the input sequence as hints for the target tasks. These prompt inputs are used to query a large language model (*e.g.*, GPT-3 [7]). Methods [25, 61] have been proposed to automate the prompt engineering process. The prompting process does not tune any of the parameters, which is empirically sub-optimal compared to fine-tuning [43].

Prompt tuning [33] is proposed to replace hard language prompts with learnable prompt tokens that can be updated through gradient back-propagation, while other parameters are still kept fixed. Other than adding learnable input tokens, Prefix-Tuning [38] adds learnable prompts to each layer of the Transformer to boost the model capacity. For few-shot scenario, [20] proves that prompt tuning can be much better than traditional fine-tuning. When the training data is sufficient, prompt tuning performs slightly worse than fine-tuning [43]. However, the performance gap from full-model fine-tuning closes up as the pre-trained model gets larger [33, 42]. Inspired by the success of prompt tuning in NLP, [77] applies prompt tuning to visual-linguistic pre-trained models (*e.g.*, CLIP [55]) to perform few-shot image classification. [51, 58] further apply a residual feature adapter to improve the few-shot performance.

In this paper, we focus on the zero-shot and few-shot scenarios, where the downstream tasks may not even appear in the pre-training stage. For few-shot learning, we adapt the model with prompt tuning proposed by [41]. The performance of our model can be further improved by fine-tuning the whole model with sufficient downstream training data.

3. Method

3.1. Unified Architecture for Generic Perception

In this section, we will describe our unified architecture for various modalities and tasks. Fig. 2 illustrates the architecture. Specifically, the model first converts different task inputs and targets from arbitrary combinations of modalities into token sequences with modality-specific tokenizers. A modality-agnostic Transformer encoder, which shares parameters for different input modalities and target tasks, is then employed to encode different token sequences into a shared representation space. Any perception task can be

在 Visual Genome [30] 上预训练的 CNN [57]）。[24] 试图通过使用 CNN 直接从原始像素中提取特征，来消除对目标检测器的需求。[1, 22, 28, 56] 进一步将 Transformer 应用于原始图像块和词元。Transformer 为不同模态提供了一种统一的架构范式，只需要特定于模态的分词器将来自不同模态的输入转换为统一的输入词元序列。

然而，以往的架构需要为不同的感知任务专门设计并训练预测头。相反，我们通过将不同任务的输入和目标编码到统一的表示空间中，并通过表示相似性来建模它们之间的关系，以取代任务特定的设计。这一修改使我们的模型即使在预训练阶段未出现过的新下游任务上，也能够进行零样本推理。

预训练。大规模预训练在深度学习领域取得了巨大成功，能够缓解数据需求量大的挑战，并提升下游任务的性能 [72]。对于图像识别，预训练通常在图像分类数据集上进行，例如 ImageNet [15]。视频识别网络则通常在图像分类或视频分类数据集上进行预训练，例如 Moments in Time [49] 和 Kinetics [27]。在自然语言处理中，自监督语言建模 [7, 16, 31, 34, 44] 被用于在大规模无标注语料库 [54] 上进行预训练。具体而言，GPT [7] 执行自回归预训练，即在给定前文词语的条件下优化下一个词的概率。BERT [16] 使用掩码语言建模（MLM）和下一句预测（NSP）进行预训练。这些预训练模型可以作为下游任务的稳健特征提取器，只需对架构进行小幅修改。

近年来，大规模跨模态预训练 [72] 受到了关注。与单模态预训练相比，跨模态预训练需要对齐来自不同模态的信息。此类预训练通常在从互联网收集的图像-文本对 [8, 26, 50, 60] 以及人工标注的视觉-语言数据集 [30, 40, 50] 上进行。此外，还提出了各种预训练目标，以有效利用这些数据集。最广泛使用的目标包括图像-文本检索 [2, 37, 47, 55, 63, 64, 65]、带有图像线索的掩码语言建模 [2, 37, 47, 62, 63, 64, 65]，以及掩码区域建模 [14, 47, 62, 63, 65]。其中，掩码区域建模需要由现成的目标检测器提取的区域特征。最近，CLIP [55] 已经验证了仅在大规模网络收集数据上进行图像-文本检索预训练的有效性。

以往的多任务预训练需要特定于任务的头部，这阻碍了不同任务之间的协作。相反，我们对不同任务的输入和目标进行编码进入统一的表示空间，并通过统一的表示相似性来建模它们之间的关系，从而实现不同模态和任务之间的协作。我们的预训练任务包括图像和视频分类、有无图像线索的语言建模，以及图像-文本检索。我们不使用区域特征及其对应的预训练任务。

提示调优。作为微调的替代方案，提示调优最近在自然语言处理社区中被提出，其起源于提示方法 [41]。在提示方法中，专门设计的自然语言标记，即提示，被插入到输入序列中，作为目标任务的线索。这些提示输入用于查询大型语言模型（例如 GPT-3 [7]）。已有方法 [25, 61] 被提出用于自动化提示工程过程。提示过程不会调优任何参数，而从经验上看，与微调相比这是次优的 [43]。

提示调优 [33] 被提出用于用可学习的提示词元替代硬语言提示，这些提示词元可以通过梯度反向传播进行更新，而其他参数仍保持固定。除了添加可学习的输入词元之外，Prefix-Tuning [38] 还在 Transformer 的每一层添加可学习提示，以提升模型容量。对于少样本场景，[20] 证明提示调优可以远优于传统微调。当训练数据充足时，提示调优的表现略逊于微调 [43]。然而，随着预训练模型规模变大，与全模型微调之间的性能差距会逐渐缩小 [33, 42]。受 NLP 中提示调优成功的启发，[77] 将提示调优应用于视觉-语言预训练模型（例如 CLIP [55]），以执行少样本图像分类。[51, 58] 进一步应用残差特征适配器来提升少样本性能。

在本文中，我们聚焦于零样本和少样本场景，其中下游任务甚至可能不会出现在预训练阶段。对于少样本学习，我们采用[41]提出的提示调优来适配模型。在拥有充足下游训练数据的情况下，通过微调整个模型，可以进一步提升我们模型的性能。

3. 方法

3.1. 通用感知的统一架构

在本节中，我们将描述用于各种模态和任务的统一架构。图 2 展示了该架构。具体而言，模型首先使用特定于模态的分词器，将不同任务的输入和目标从任意模态组合转换为词元序列。然后，采用一个模态无关的 Transformer 编码器，将不同的词元序列编码到共享表示空间中；该编码器在不同输入模态和目标任务之间共享参数。任何感知任务都可以

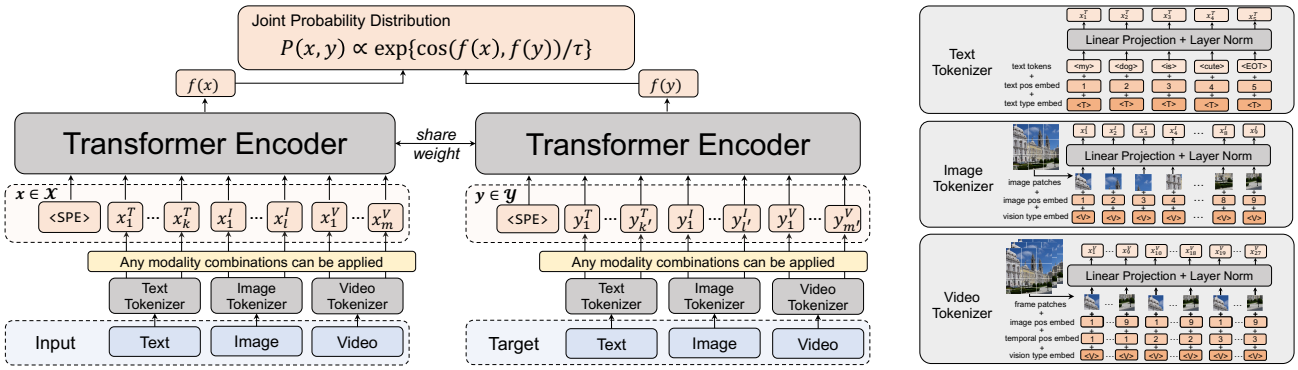


Figure 2. Overview of our unified architecture for generic perception. Different task inputs and targets from arbitrary modalities are converted into unified token sequences with modality-specific tokenizers. A modality-agnostic weight-sharing Transformer encoder is then applied to encode these token sequences into the shared representation space. Any perception task can be modeled as finding the maximum likelihood target for each input through the similarity of their representations.

modeled in a single unified formulation, which finds the maximum likelihood target for each input through the similarity of their representations.

Tokenization. Given the raw inputs from text, image, and video modalities, modality-specific tokenizers are applied to generate the input token sequences for the Transformer encoder. Here, we use the BPE tokenizer [59] for text modality, the image patch tokenizer [17] for image modality, and the temporal frame patch tokenizer [6] for video modality. These outputted tokens are attached with additional modality type embeddings to identify which modality the raw input belongs to. Details of the modality-specific tokenizers are described in the Appendix.

As illustrated in Fig. 2, depending on the task requirements, the input sequence x of the Transformer encoder can be composed of different combinations of text token sequence x^T, image token sequence x^I, and video token sequence x^V. At the beginning of the sequence x, a special token <SPE> is always inserted. For example, $x = [\texttt{<SPE>}, x^I, x^T]$ for image-text pair inputs, and $x = [\texttt{<SPE>}, x^V]$ for video-only inputs, where $[\,]$ denotes the sequence concatenation. The feature of <SPE> at the encoder output serves as the representation of the input.

Generic Modeling of Perception Tasks. We model different perception tasks with a unified architecture, whose parameters are shared for all target tasks. Each task is defined with a set of inputs $\mathcal{X}$ and a set of candidate targets $\mathcal{Y}$. Given an input $x \in \mathcal{X}$, the task is formulated as finding the maximum likelihood target $y \in \mathcal{Y}$ as

$$\hat{y} = \arg\max_{y \in \mathcal{Y}} P(x, y), \tag{1}$$

where $P(x, y)$ is the joint probability distribution. The joint probability is estimated through calculating the cosine similarity between the representation of x and y as

$$P(x, y) \propto \exp\left(\cos\left(f(x), f(y)\right)/\tau\right), \tag{2}$$

where $f(\cdot)$ is the Transformer encoder, and $\tau > 0$ is a learnable temperature parameter.

To obtain generic modeling capability, our unified architecture is pre-trained on a variety of multi-modal tasks simultaneously. Suppose a series of pre-training tasks is denoted as $\{\mathcal{X}_1, \mathcal{Y}_1\}, \{\mathcal{X}_2, \mathcal{Y}_2\}, ..., \{\mathcal{X}_n, \mathcal{Y}_n\}$, where $\mathcal{X}_i$ and $\mathcal{Y}_i$ is the input set and target set of the i-th task, respectively. Then the pre-training loss is defined as

$$L = \sum_{i=1}^{n} \mathop{\mathbb{E}}_{\{x,y\} \in \{\mathcal{X}_i, \mathcal{Y}_i\}} \left[-\log \frac{P(x, y)}{\sum_{z \in \mathcal{Y}_i} P(x, z)} \right], \tag{3}$$

where $\mathbb{E}$ is the mathematical expectation, and $\{x, y\} \in \{\mathcal{X}_i, \mathcal{Y}_i\}$ indicates a ground-truth input-target pair sampled from the dataset of the i-th task.

Our unified architecture is suitable for any task, as long as its input set $\mathcal{X}$ and target set $\mathcal{Y}$ are composed of images, texts, and videos. For example, the target set $\mathcal{Y}$ in classification tasks can be a set of class names, a set of class descriptions, or even a set of images with handwritten numbers representing class indexes. Detailed instances of $\mathcal{X}$ and $\mathcal{Y}$ will be introduced in the next subsection. Note that we currently focus on text, image, and video modalities, but more modalities are also applicable, as long as the corresponding tokenizers are applied.

Relation to Previous Perception Models. Our method shares the same goal of learning multi-modal representations as previous perception models. However, existing works follow a task-specific paradigm, while our method is designed for generic perceptual modeling. The main difference lies in two parts:

1) Previous works focus on inputs from certain combinations of modalities required by their target tasks, while our

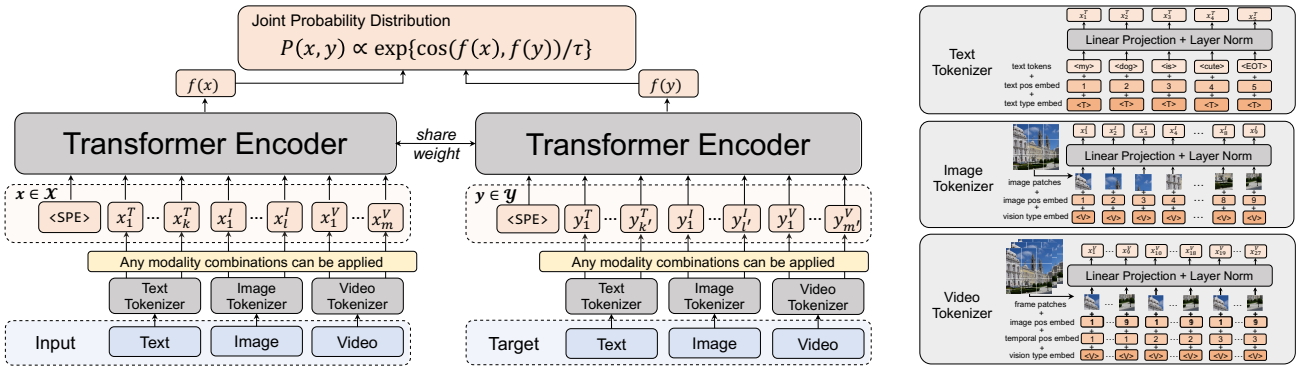


图 2. 我们用于通用感知的统一架构概览。来自任意模态的不同任务输入和目标通过特定于模态的分词器转换为统一的标记序列。随后，采用模态无关的权重共享 Transformer 编码器，将这些标记序列编码到共享表示空间中。任何感知任务都可以建模为：通过输入与目标表示之间的相似性，为每个输入寻找最大似然目标。

在一个统一的公式中建模，该公式通过其表示之间的相似性为每个输入找到最大似然目标。

分词。给定来自文本、图像和视频模态的原始输入，应用特定于模态的分词器，为 Transformer 编码器生成输入词元序列。在这里，我们对文本模态使用 BPE 分词器 [59]，对图像模态使用图像块分词器 [17]，对视频模态使用时间帧块分词器 [6]。这些输出的词元会附加额外的模态类型嵌入，以标识原始输入属于哪种模态。特定于模态的分词器的详细信息见附录。

如图 2 所示，根据任务需求，Transformer 编码器的输入序列 x 可以由文本 token 序列 x^T、图像 token 序列 x^I 和视频 token 序列 x^V 的不同组合构成。在序列 x 的开头，始终会插入一个特殊 token <SPE>。例如，对于图文对输入为 x = [<SPE>, x^I, x^T]，对于仅视频输入为 x = [<SPE>, x^V]，其中 [ ] 表示序列拼接。编码器输出中 <SPE> 的特征作为输入的表示。

感知任务的通用建模。我们使用统一架构对不同的感知任务进行建模，其参数在所有目标任务之间共享。每个任务由一组输入 $\mathcal{X}$ 和一组候选目标 $\mathcal{Y}$ 定义。给定输入 $x \in \mathcal{X}$，该任务被表述为寻找最大似然目标 $y \in \mathcal{Y}$，即

$$\hat{y} = \arg\max_{y \in \mathcal{Y}} P(x, y), \tag{1}$$

其中 $P(x, y)$ 是联合概率分布。联合概率通过计算余弦相似度来估计。

x 和 y 的表示之间的相似性为

$$P(x, y) \propto \exp\left(\cos\left(f(x), f(y)\right)/\tau\right), \tag{2}$$

其中，$f(\cdot)$ 是 Transformer 编码器，$\tau > 0$ 是一个可学习的温度参数。

为了获得通用的建模能力，我们的统一架构会同时在多种多模态任务上进行预训练。假设一系列预训练任务表示为 $\{\mathcal{X}_1, \mathcal{Y}_1\}, \{\mathcal{X}_2, \mathcal{Y}_2\}, ..., \{\mathcal{X}_n, \mathcal{Y}_n\}$，其中 $\mathcal{X}_i$ 和 $\mathcal{Y}_i$ 分别是第 i 个任务的输入集和目标集。那么预训练损失定义为：

$$L = \sum_{i=1}^{n} \mathop{\mathbb{E}}_{\{x,y\} \in \{\mathcal{X}_i, \mathcal{Y}_i\}} \left[-\log \frac{P(x, y)}{\sum_{z \in \mathcal{Y}_i} P(x, z)} \right], \tag{3}$$

其中，$\mathbb{E}$ 表示数学期望，$\{x, y\} \in \{\mathcal{X}_i, \mathcal{Y}_i\}$ 表示从第 i 个任务的数据集中采样得到的真实输入-目标对。

我们的统一架构适用于任何任务，只要其输入集 $\mathcal{X}$ 和目标集 $\mathcal{Y}$ 由图像、文本和视频组成。例如，分类任务中的目标集 $\mathcal{Y}$ 可以是一组类别名称、一组类别描述，甚至可以是一组带有手写数字、表示类别索引的图像。$\mathcal{X}$ 和 $\mathcal{Y}$ 的详细实例将在下一小节中介绍。请注意，我们目前关注文本、图像和视频模态，但只要应用相应的分词器，更多模态也同样适用。

与以往感知模型的关系。我们的方法与以往的感知模型具有相同的目标，即学习多模态表征。然而，现有工作遵循特定任务的范式，而我们的方法旨在进行通用感知建模。主要区别体现在两个方面：

1）以往的工作关注其目标任务所需的某些模态组合的输入，而我们的

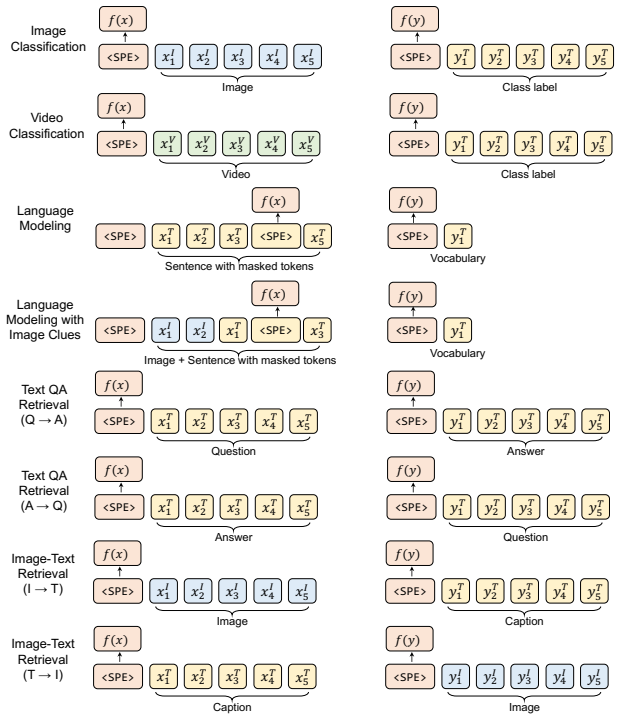


Figure 3. Input and target formats of pre-training tasks. For each task, the left column represents the format of input sequence x, and the right column represents the format of the target sequence y. $f(x)$ and $f(y)$ indicate the representations used for calculating the joint probability distribution as in Eq. (2). Here, we have omitted the tokenizer and encoder for concision.

method handles arbitrary combinations of modalities with a unified architecture and shared parameters.

2) Previous works require prediction heads specifically designed and trained for each perception task, while our method models different tasks with the same formulation and processes them with unified modeling.

Therefore, when transferred to a new task, previous methods need to re-design their input formats and prediction heads accordingly. Models require fine-tuning on sufficient task-specific data, resulting in remarkable human and computational costs. In contrast, our method can directly conduct zero-shot inference on novel tasks that do not appear in the pre-training stage. The performance can be further boosted with prompt tuning on few-shot downstream data and fine-tuning on sufficient downstream data.

3.2. Pre-training on Multi-Modal Tasks

Our model is pre-trained on a variety of tasks simultaneously to learn the multi-modal generic representations. The pre-training tasks are illustrated in Fig. 3. Specifically, for uni-modal pre-training tasks, we adopt the most widely-used image classification, video classification, and language modeling tasks. To further enhance the relationships between different modalities, some cross-modal tasks are also employed, such as language modeling with image clues and image-text retrieval tasks. Note that for image and video classification tasks, we regard each class name (*e.g.*, `tigershark`) as a text sequence. This provides weak supervision for bridging the gap among the representations of images, videos, and texts.

Image and Video Classification. In image and video classification tasks, $\mathcal{X}$ denotes the set of all possible images or videos in the training dataset, and $\mathcal{Y}$ consists of candidate class labels in each dataset. Each class name is regarded as a text sequence to provide weak supervision of the relationship to texts. Both the input $x \in \mathcal{X}$ and target $y \in \mathcal{Y}$ start with an <SPE> token, whose feature at the encoder output represents the corresponding sequence.

Language Modeling with and without Image Clues. The language modeling task aims to predict the masked words according to the context. Both auto-regressive [7] and auto-encoding [16] language modeling are adopted. When inputs have no image, the auto-regressive and auto-encoding tasks correspond to the text generation and the masked language modeling tasks, respectively. When inputs have images, the auto-regressive and auto-encoding tasks correspond to the image caption and the masked language modeling with image clues tasks, respectively.

For auto-encoding language modeling, we follow the practice in BERT [16] to mask out 15% words from the text randomly. The model predicts each masked word based on all inputs. For auto-regressive language modeling, the model predicts each word based on its previous text and image (if any). Please refer to the Appendix for an efficient implementation of auto-regressive language modeling.

In this task, $\mathcal{X}$ consists of language sentences or image-text pairs. $\mathcal{Y}$ denotes the set of all words in the vocabulary, where each word is regarded as a single text sequence. Each word that needs to be predicted in $x \in \mathcal{X}$ is replaced by a <SPE> token, whose feature at the encoder output is used to match the words in the vocabulary $\mathcal{Y}$.

Image and Text Retrieval. For image-text retrieval, the input sets $\mathcal{X}$ and $\mathcal{Y}$ are composed of images and text sequences respectively, or vice versa. For text-only retrieval, the input sets $\mathcal{X}$ and $\mathcal{Y}$ are both text sequences. Each sequence in $\mathcal{X}$ and $\mathcal{Y}$ has a special token <SPE> at the beginning, whose feature at the output of the encoder serves as the final representation.

3.3. Zero-shot, Prompt Tuning and Fine-tuning

During the pre-training stage, our unified architecture learns to model the joint distribution of input and target sequences from arbitrary modalities. Thanks to the generic perceptual modeling, our pre-trained model can perform zero-shot inference on completely novel tasks that do not appear in the pre-training stage. Our model can be further

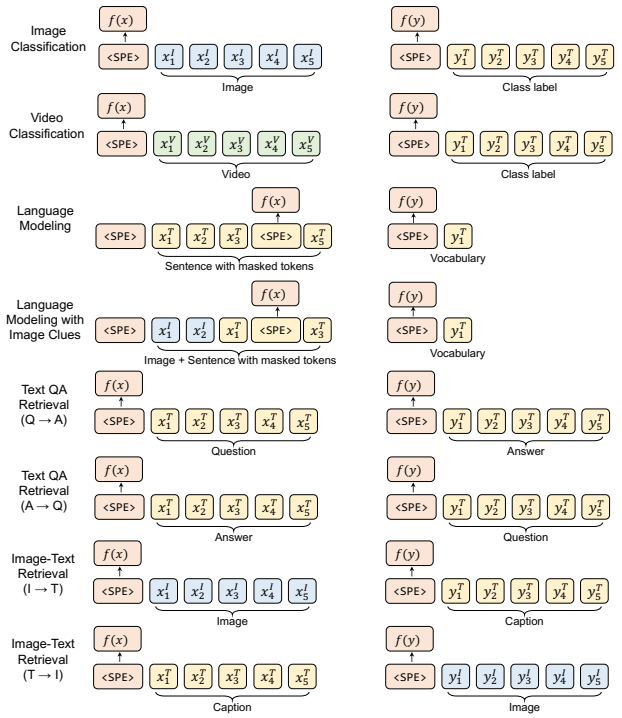


图3. 预训练任务的输入和目标格式。对于每个任务，左列表示输入序列 x 的格式，右列表示目标序列 y 的格式。$f(x)$ 和 $f(y)$ 表示用于计算公式（2）中的联合概率分布的表示。为简洁起见，此处省略了分词器和编码器。

该方法通过统一的架构和共享参数处理任意模态组合。

2) 以往工作需要为每个感知任务专门设计并训练预测头，而我们的方法用相同的表述对不同任务进行建模，并以统一建模的方式处理它们。

因此，当迁移到新任务时，以往的方法需要相应地重新设计其输入格式和预测头。模型需要在充足的任务特定数据上进行微调，从而带来显著的人力和计算成本。相比之下，我们的方法可以直接在预训练阶段未出现的新任务上进行零样本推理。通过在少样本下游数据上进行提示调优，以及在充足的下游数据上进行微调，性能还可以进一步提升。

3.2. 多模态任务预训练

我们的模型在多种任务上同时进行预训练，以学习多模态通用表示。预训练任务如图 3 所示。具体而言，对于单模态预训练任务，我们采用最广泛使用的图像分类、视频分类和语言建模任务。为了进一步增强不同模态之间的关系，一些跨模态任务还采用了其他任务，例如基于图像线索的语言建模和图文检索任务。请注意，对于图像和视频分类任务，我们将每个类别名称（例如，tigershark）视为一个文本序列。这为弥合图像、视频和文本表示之间的差距提供了弱监督。

图像和视频分类。在图像和视频分类任务中，$\mathcal{X}$ 表示训练数据集中所有可能的图像或视频的集合，$\mathcal{Y}$ 由每个数据集中的候选类别标签组成。每个类别名称都被视为一个文本序列，以提供与文本关系的弱监督。输入 $x \in \mathcal{X}$ 和目标 $y \in \mathcal{Y}$ 都以一个 <SPE> 标记开头，其在编码器输出处的特征表示相应的序列。

有无图像线索的语言建模。语言建模任务旨在根据上下文预测被遮蔽的词。采用自回归 [7] 和自编码 [16] 两种语言建模方式。当输入不包含图像时，自回归任务和自编码任务分别对应文本生成任务和掩码语言建模任务。当输入包含图像时，自回归任务和自编码任务分别对应图像描述任务和带有图像线索的掩码语言建模任务。

对于自编码语言建模，我们遵循 BERT [16] 中的做法，从文本中随机掩蔽 15% 的词。模型基于所有输入预测每个被掩蔽的词。对于自回归语言建模，模型基于其之前的文本和图像（如果有）预测每个词。有关自回归语言建模的高效实现，请参阅附录。

在此任务中，$\mathcal{X}$由语言句子或图像-文本对组成。$\mathcal{Y}$ 表示词汇表中所有词的集合，其中每个词都被视为一个单独的文本序列。$x \in \mathcal{X}$中需要预测的每个词都由一个<SPE>标记替换，其在编码器输出处的特征用于匹配词汇表$\mathcal{Y}$中的词。

图像和文本检索。对于图像-文本检索，输入集合 $\mathcal{X}$ 和 $\mathcal{Y}$ 分别由图像和文本序列组成，反之亦然。对于纯文本检索，输入集合 $\mathcal{X}$ 和 $\mathcal{Y}$ 均为文本序列。$\mathcal{X}$ 和 $\mathcal{Y}$ 中的每个序列在开头都有一个特殊标记 <SPE>，其在编码器输出处的特征作为最终表示。

3.3. 零样本、提示调优和微调

在预训练阶段，我们的统一架构学习对来自任意模态的输入序列和目标序列的联合分布进行建模。得益于通用的感知建模，我们的预训练模型能够在预训练阶段未出现的全新任务上进行零样本推理。我们的模型还可以进一步

adapted to downstream tasks with task-specific additional training data. For the few-shot scenario, we employ the prompt tuning [41] scheme, which only adds a few additional task-specific parameters to the model. The performance on specific tasks can be further improved by fine-tuning the whole model on sufficient downstream data.

Zero-shot Inference on Novel Tasks. Our model has the potential to perform zero-shot inference on any perception task that can be modeled by a joint probability distribution. For a task with input $x \in \mathcal{X}$ and a candidate target $y \in \mathcal{Y}$, we firstly tokenize x and y into two sequences. The joint probability $P(x, y)$ is then estimated following Eq. (2). Zero-shot inference can be conducted by maximum likelihood estimation, as described in Eq. (1). Performance can also be improved through prompt engineering, similar to the prompting [41] for language models such as GPT-3 [7], where network training is not required.

Prompt Tuning. For the few-shot scenario with limited training data, we adopt prompt tuning, which is memory-efficient and has been proved to be better than the fine-tuning scheme in few-shot NLP [20]. In prompt tuning, most pre-trained parameters are fixed, leaving only a small portion of task-specific parameters to be optimized. Specifically, following P-Tuning v2 [42], learnable prompt tokens with random initialization are added at each layer of the Transformer encoder, and class labels with linear heads are added for classification tasks. The <SPE> token and layer norm parameters are also tuned. We refer the readers to the Appendix for more details.

Fine-Tuning. For downstream tasks with sufficient training data, our model can also be fine-tuned to further improve its performance. During fine-tuning, our model can serve as a joint probability estimator (same as our proposed generic perceptual modeling), or a feature extractor (same as traditional pre-trained models). Under the setting of joint probability estimation, the downstream tasks are formulated in the same unified manner as in pre-training. On the other hand, similar to previous perception models, our model can also be used as a feature extractor by adding a task-specific head on the top of the encoder. We empirically find these two schemes achieve very similar performance, and hence the scheme of joint probability distribution estimator is used by default for consistency.

4. Experiments

4.1. Datasets

Our model is pre-trained on a variety of tasks, whose statistics are listed in Tab. 1. We pre-train image classification on ImageNet-21k [15]. For video classification, we pre-train on Kinetics-700 [27] and Moments in Time [49]. We pre-train language modeling on BookCorpora [79] & English Wikipedia (Books&Wiki) and PAQ [35]. For language modeling with image clues and image-text retrieval, we use a combination of COCO Caption [12], SBU Captions (SBU) [50], Visual Genome [30], CC3M [60], CC12M [8] and YFCC [26]. To evaluate the effectiveness of our method and verify the generalization of our pre-trained model, we also use several novel datasets that did not appear in pre-training, *i.e.*, Flickr30k [52], MSVD [10], VQA [21], and GLUE [70]. See Appendix for the details of datasets.

Dataset	#Images	#Videos	#Text
ImageNet-21k [15]	14.2M	0	21K
Kinetics-700 [27]	0	542K	700
Moments in Time [49]	0	792K	339
Books&Wiki [79]	0	0	101M
PAQ [35]	0	0	65M
CC3M [60]	3.0M	0	3.0M
CC12M [8]	11.1M	0	11.1M
COCO Caption [11]	113K	0	567K
Visual Genome [30]	108K	0	5.41M
SBU [50]	830K	0	830K
YFCC* [26]	14.8M	0	14.8M

Table 1. Pre-training dataset statistics. #Images, #Videos and #Text represent the number of images, video clips, and textual sentences (or phrases), respectively.

4.2. Implementation Details

The Transformer encoder used for experiments is of the same configuration with $\text{BERT}_{\text{BASE}}$ [16]. It is a 12-layer encoder with the embedding dimension of 768 and the attention head number of 12. The hidden dimension size in FFN is 3072. We pre-train the model with multiple tasks simultaneously. In each iteration, each GPU independently samples a single task and dataset. The gradients of different GPUs are synchronized after the gradient back-propagation. We use AdamW [29] optimizer with a base learning rate of 0.0002 and a weight decay of 0.05. Gradient clipping with 5.0 is used to stabilize training. We also use drop path [32] with a probability of 0.1 during training. The model is pre-trained on 128 Tesla V100 GPUs in a distributed fashion for 500k iterations. We use the cosine learning rate schedule with 50k iterations of linear warmup. See Appendix for more implementation details.

4.3. Evaluation on Pre-training Tasks

We first evaluate our pre-trained model on tasks that have been involved in the pre-training stage, while the datasets might be different. The widely used Imagenet-1k [15] and Kinetics-400 [27] are used for evaluating the image and video classification tasks, respectively. COCO Caption and Flickr30k are the typical datasets used to evaluate the performance on image caption and image-text retrieval.

Results. Tab. 2, Tab. 3, and Tab. 4 present the evaluation results of our models on four pre-training tasks, *i.e.*, image

通过使用特定任务的额外训练数据适配下游任务。对于少样本场景，我们采用提示调优 [41] 方案，该方案仅向模型添加少量额外的特定任务参数。在充足的下游数据上微调整个模型，可以进一步提升其在特定任务上的性能。

对新任务的零样本推理。我们的模型有潜力在任何可以由联合概率分布建模的感知任务上执行零样本推理。对于输入为 $x \in \mathcal{X}$、候选目标为 $y \in \mathcal{Y}$ 的任务，我们首先将 x 和 y 标记化为两个序列。然后按照式（2）估计联合概率 $P(x, y)$。零样本推理可以通过最大似然估计来进行，如式（1）所述。性能也可以通过提示工程来提升，类似于针对 GPT-3 [7] 等语言模型的提示方法 [41]，且无需进行网络训练。

提示调优。对于训练数据有限的少样本场景，我们采用提示调优，这种方法具有较高的内存效率，并已被证明在少样本 NLP 中优于微调方案 [20]。在提示调优中，大多数预训练参数保持固定，仅优化一小部分任务特定参数。具体而言，遵循 P-Tuning v2 [42]，在 Transformer 编码器的每一层添加随机初始化的可学习提示标记，并为分类任务添加带有线性头的类别标签。<SPE> 标记和层归一化参数也会被调优。更多细节请读者参阅附录。

微调。对于具有足够训练数据的下游任务，我们的模型也可以进行微调，以进一步提升其性能。在微调过程中，我们的模型可以充当联合概率估计器（与我们提出的通用感知建模相同），也可以充当特征提取器（与传统的预训练模型相同）。在联合概率估计的设置下，下游任务以与预训练相同的统一方式进行表述。另一方面，类似于以往的感知模型，我们的模型也可以通过在编码器顶部添加特定任务的头部来用作特征提取器。我们通过实验证明，这两种方案取得了非常相近的性能，因此为保持一致性，默认采用联合概率分布估计器的方案。

4. 实验

4.1. 数据集

我们的模型在多种任务上进行预训练，其统计信息列于表 1。我们在 ImageNet-21k [15] 上预训练图像分类。对于视频分类，我们在 Kinetics-700 [27] 和 Moments in Time [49] 上进行预训练。我们在 BookCorpora [79] & 上预训练语言建模。

Dataset	#Images	#Videos	#Text
ImageNet-21k [15]	14.2M	0	21K
Kinetics-700 [27]	0	542K	700
Moments in Time [49]	0	792K	339
Books&Wiki [79]	0	0	101M
PAQ [35]	0	0	65M
CC3M [60]	3.0M	0	3.0M
CC12M [8]	11.1M	0	11.1M
COCO Caption [11]	113K	0	567K
Visual Genome [30]	108K	0	5.41M
SBU [50]	830K	0	830K
YFCC* [26]	14.8M	0	14.8M

表 1. 预训练数据集统计。#Images、#Videos 和 #Text 分别表示图像、视频片段和文本句子（或短语）的数量。

英文维基百科（Books&Wiki）和 PAQ [35]。对于带有图像线索的语言建模和图像-文本检索，我们使用 COCO Caption [12]、SBU Captions（SBU）[50]、Visual Genome [30]、CC3M [60]、CC12M [8] 和 YFCC [26] 的组合。为评估我们方法的有效性并验证预训练模型的泛化能力，我们还使用了若干未出现在预训练中的新数据集，即 Flickr30k [52]、MSVD [10]、VQA [21] 和 GLUE [70]。数据集的详细信息见附录。

4.2. 实现细节

用于实验的 Transformer 编码器配置与 $\text{BERT}_{\text{BASE}}$ [16] 相同。它是一个 12 层编码器，嵌入维度为 768，注意力头数为 12。FFN 中的隐藏维度大小为 3072。我们同时使用多项任务对模型进行预训练。在每次迭代中，每个 GPU 独立采样一个任务和数据集。在梯度反向传播后，同步不同 GPU 的梯度。我们使用 AdamW [29] 优化器，基础学习率为 0.0002，权重衰减为 0.05。采用 5.0 的梯度裁剪来稳定训练。训练期间我们还使用概率为 0.1 的 drop path [32]。该模型以分布式方式在 128 块 Tesla V100 GPU 上预训练 500k 次迭代。我们使用余弦学习率调度，并进行 50k 次迭代的线性预热。更多实现细节见附录。

4.3. 预训练任务评估

我们首先在预训练阶段涉及的任务上评估我们的预训练模型，尽管数据集可能不同。广泛使用的 Imagenet-1k [15] 和 Kinetics-400 [27] 分别用于评估图像和视频分类任务。COCO Caption 和 Flickr30k 是用于评估图像描述和图像-文本检索性能的典型数据集。

结果。表2、表3和表4展示了我们的模型在四项预训练任务上的评估结果，即图像

Task	ImageNet-1k Acc	Kinetics-400 Acc
DeiT [67]	81.8	-
TimeSformer [6]	-	75.5
Ours $_{\text{w/o Tuning}}$	78.0	73.5
Ours $_{\text{PT (0.1\%)}}$	79.4	73.6
Ours $_{\text{FT (0.1\%)}}$	78.8	73.5
Ours $_{\text{PT (1\%)}}$	80.2	73.6
Ours $_{\text{FT (1\%)}}$	80.2	73.6
Ours $_{\text{FT (100\%)}}$	83.8	75.8

Table 2. Image and video classification performance under different tuning settings. PT means prompt-tuning, and FT means fine-tuning. The percentage of data used in tuning is noted.

classification, video classification, image-text retrieval, and image caption. We compare our model with task-specific SOTA methods having the similar model size.

Results show that without any tuning, our pre-trained model reaches reasonable performance on these tasks. Although the performance is slightly worse than the SOTA methods. We speculate that the performance gap is due to the limited capacity of our model, which may have a negative impact on the representation ability. Note that our method shares a similar model size with other methods, but need to simultaneously process much more pre-training tasks from various datasets and modalities.

By conducting prompt tuning on each task with 1% downstream data, the performance is boosted to a level close to SOTA performance. It's worth noting that all parameters of other methods are specifically trained on the target tasks. While for our prompt tuning, only a small amount of parameters are tuned, and the encoder is still fixed and shared among different tasks, indicating that our method can handle different tasks with low marginal cost.

We further fine-tune the pre-trained model with 100% of the downstream data. With full-data fine-tuning, our model achieves performance on-par with or better than the SOTA methods on all these tasks, which proves our model has learned high-quality representations. We also compare the performance of prompt tuning and fine-tuning in the scenario of few-shot learning. On all of these tasks, prompt tuning shows a consistently better performance than fine-tuning with the same amount of data, which demonstrates its superiority under few-shot scenarios.

4.4. Generalization to Novel Tasks

Thanks to the generic perceptual modeling, our pre-trained model can generalize to novel tasks by converting the tasks into our unified task formulation. We evaluate zero-shot inference on tasks that did not appear in pre-training, *i.e.*, video caption, video-text retrieval, visual question answering, and natural language understanding.

Video Caption and Video-Text Retrieval. Our pre-trained model is evaluated on MSVD [10] dataset. Specifically, for video caption, $\mathcal{X}_1$ consists of the concatenation of video and language sequences that have been predicted, and $\mathcal{X}_2$ denotes the set of all words in the vocabulary. For the video-text retrieval, the input sets $\mathcal{X}_1$ and $\mathcal{X}_2$ consist of possible video and text sequences, or vice versa.

Visual Question Answering. In visual question answering, the model is asked to answer a question w.r.t a reference image from a list of answer candidates. We evaluate our pre-trained model on VQA [21] dataset. $\mathcal{X}_1$ is a set of image-text sequence, where the text is the question tokens followed by a <SPE> token used to predict the answers. Each $x_2 \in \mathcal{X}_2$ is an answer sequence beginning with <SPE>. Inference is achieved by computing the similarity between output features of <SPE> in x_1 and x_2.

Natural Language Understanding. Six language-only tasks are chosen from GLUE benchmark [70] to evaluate the natural language understanding ability of our pre-trained model. These tasks are either single sentence classification or sentence-pair classification tasks. We follow [19] to construct the textual class labels for each dataset. Here, the input sequence $x_1 \in \mathcal{X}_1$ denotes the input single sentence or the sentence-pair, and the sequence $x_2 \in \mathcal{X}_2$ represents the class labels in each dataset.

Result. Tab. 5, Tab. 6 and Tab. 7 show the results on video caption and video-text retrieval and visual question answering, respectively. Our pre-trained model can obtain reasonable zero-shot performance on these novel tasks. Note that none of previous works can perform this type of zero-shot inference at all. From Tab. 7, we note that our model shows unsatisfactory zero-shot performance on "Yes/No" and "Number" subsets in VQA. We speculate that it may be due to the distribution difference between those answers and our pre-training corpora. We futher conduct prompt tuning on these tasks with only 1% data, which brings our model to a level close to the SOTA results. By further fine-tuning with 100% downstream data, our model can achieve results on par with or better than the SOTA methods.

On the GLUE benchmark, our model can achieve comparable performance with [3] in zero-shot evaluation. When fine-tuning the pre-trained model with 100% downstream data, our model performs slightly worse than $\text{BERT}_{\text{BASE}}$. Since our model has the same number of parameters as $\text{BERT}_{\text{BASE}}$, but need to process much more tasks from various datasets and modalities, we speculate that the performance drop is due to the limited capacity of the model.

5. Conclusion

In this paper, we propose a unified perception architecture that processes various modalities and tasks with a single model and shared parameters. With pre-training on uni-modal and multi-modal tasks, our model shows the ability

Task	ImageNet-1k Acc	Kinetics-400 Acc
DeiT [67]	81.8	-
TimeSformer [6]	-	75.5
Ours $_{\text{w/o Tuning}}$	78.0	73.5
Ours $_{\text{PT (0.1\%)}}$	79.4	73.6
Ours $_{\text{FT (0.1\%)}}$	78.8	73.5
Ours $_{\text{PT (1\%)}}$	80.2	73.6
Ours $_{\text{FT (1\%)}}$	80.2	73.6
Ours $_{\text{FT (100\%)}}$	83.8	75.8

表 2. 不同调优设置下的图像和视频分类性能。PT 表示提示调优，FT 表示微调。标注了调优中使用的数据比例。

分类、视频分类、图文检索和图像描述生成。我们将我们的模型与具有相似模型规模的特定任务 SOTA 方法进行比较。

结果表明，在没有任何调优的情况下，我们的预训练模型在这些任务上达到了合理的性能。尽管其性能略逊于 SOTA 方法。我们推测，性能差距源于我们模型容量有限，这可能会对表征能力产生负面影响。需要注意的是，我们的方法与其他方法具有相近的模型规模，但需要同时处理来自各种数据集和模态的更多预训练任务。

通过在每个任务上使用 1% 的下游数据进行提示调优，性能提升到了接近 SOTA 的水平。值得注意的是，其他方法的所有参数都是针对目标任务专门训练的。而对于我们的提示调优，只需调优少量参数，编码器仍然保持固定并在不同任务之间共享，这表明我们的方法能够以较低的边际成本处理不同任务。

我们进一步使用 100% 的下游数据对预训练模型进行微调。在全量数据微调下，我们的模型在所有这些任务上的性能达到或优于 SOTA 方法，这证明我们的模型学习到了高质量的表示。我们还比较了在少样本学习场景下提示调优和微调的性能。在所有这些任务中，提示调优在相同数据量下始终表现优于微调，这表明其在少样本场景下的优越性。

4.4. 泛化到新任务

得益于通用的感知建模，我们的预训练模型可以通过将任务转换为统一的任务表述来泛化到新任务。我们在预训练中未出现的任务上评估零样本推理，即视频字幕生成、视频-文本检索、视觉问答和自然语言理解。

视频字幕和视频-文本检索。我们预训练的模型在 MSVD [10] 数据集上进行评估。具体而言，对于视频字幕生成，$\mathcal{X}_1$ 由已预测的视频序列和语言序列的拼接组成，$\mathcal{X}_2$ 表示词汇表中所有单词的集合。对于视频-文本检索，输入集合 $\mathcal{X}_1$ 和 $\mathcal{X}_2$ 由可能的视频序列和文本序列组成，反之亦然。

视觉问答。在视觉问答中，模型需要针对一张参考图像，从候选答案列表中回答一个问题。我们在 VQA [21] 数据集上评估预训练模型。$\mathcal{X}_1$ 是一组图像-文本序列，其中文本为问题词元，后接一个用于预测答案的 <SPE> 标记。每个 $x_2 \in \mathcal{X}_2$ 都是一个以 <SPE> 开头的答案序列。推理通过计算 x_1 和 x_2 中 <SPE> 的输出特征之间的相似度来实现。

自然语言理解。我们从 GLUE 基准 [70] 中选择了六个仅语言任务，以评估我们预训练模型的自然语言理解能力。这些任务要么是单句分类任务，要么是句子对分类任务。我们遵循 [19] 为每个数据集构建文本类别标签。这里，输入序列 $x_1 \in \mathcal{X}_1$ 表示输入的单个句子或句子对，序列 $x_2 \in \mathcal{X}_2$ 表示每个数据集中的类别标签。

结果。表 5、表 6 和表 7 分别展示了视频字幕生成、视频-文本检索和视觉问答的结果。我们的预训练模型在这些新任务上能够获得合理的零样本性能。需要注意的是，以往的工作完全无法进行这种类型的零样本推理。从表 7 可以看到，我们的模型在 VQA 的“Yes/No”和“Number”子集上的零样本性能并不理想。我们推测，这可能是由于这些答案与我们的预训练语料之间存在分布差异。我们进一步在这些任务上仅使用 1% 的数据进行提示调优，这使我们的模型达到了接近 SOTA 结果的水平。通过进一步使用 100% 的下游数据进行微调，我们的模型可以取得与 SOTA 方法相当或更优的结果。

在 GLUE 基准测试中，我们的模型在零样本评估中能够达到与 [3] 相当的性能。当使用 100% 的下游数据对预训练模型进行微调时，我们的模型表现略逊于 $\text{BERT}_{\text{BASE}}$。由于我们的模型与 $\text{BERT}_{\text{BASE}}$ 具有相同数量的参数，但需要处理来自各种数据集和模态的更多任务，我们推测性能下降是由于模型容量有限所致。

5. 结论

在本文中，我们提出了一种统一的感知架构，该架构使用单一模型和共享参数处理各种模态和任务。通过在单模态和多模态任务上进行预训练，我们的模型展现出了能力

Task	Text Retrieval						Image Retrieval					
	Flickr30k			COCO Caption			Flickr30k			COCO Caption		
	R@1	R@5	R10	R@1	R@5	R10	R@1	R@5	R10	R@1	R@5	R10
ImageBERT [53] w/o Tuning	70.7	90.2	94.0	44.0	71.2	80.4	54.3	79.6	87.5	32.3	59.0	70.2
UNITER-B [14] w/o Tuning	80.7	95.7	98.0	-	-	-	66.2	88.4	92.9	-	-	-
ViLT [28] w/o Tuning	73.2	93.6	96.5	56.5	82.6	89.6	55.0	82.5	89.8	40.4	70.0	81.1
Unicoder-VL [23]	86.2	96.3	99.0	62.3	87.1	92.8	71.5	91.2	95.2	48.4	76.7	85.9
UNITER-B	85.9	97.1	98.8	64.4	87.4	93.1	72.5	92.4	96.1	50.3	78.5	87.2
ViLT	83.5	96.7	98.6	61.5	86.3	92.7	64.4	88.7	93.8	42.7	72.9	83.1
Ours w/o Tuning	74.8	94.8	98.2	57.7	85.6	92.3	65.8	88.8	93.6	46.3	75.0	84.0
Ours PT (1%)	84.4	97.8	99.2	61.4	86.7	93.2	71.1	91.6	95.1	47.0	75.3	84.3
Ours FT (1%)	78.4	95.7	97.8	60.2	85.1	90.6	61.0	85.7	91.0	43.6	70.9	80.5
Ours PT (10%)	86.4	98.2	99.5	61.6	87.0	93.2	72.5	92.3	95.7	47.2	75.4	84.3
Ours FT (10%)	84.9	97.4	98.3	60.9	85.5	92.1	67.9	89.4	92.9	45.6	73.4	82.6
Ours FT (100%)	87.9	98.2	99.1	64.7	87.8	93.7	74.9	93.5	96.0	48.3	75.9	84.5

Table 3. Image-text retrieval performance under different tuning settings. PT means prompt-tuning, and FT means fine-tuning. The percentage of data used in tuning is noted.

Task	COCO Caption				Flickr30k			
	B@4	M	C	S	B@4	M	C	S
Unified VLP [78]	36.5	28.4	116.9	21.2	30.1	23.0	67.4	17.0
Ours w/o Tuning	33.6	27.0	109.8	20.3	17.0	16.2	41.2	11.2
Ours PT (1%)	34.3	27.2	109.6	21.2	28.1	21.6	59.1	15.6
Ours FT (1%)	28.0	26.8	100.1	20.2	18.9	19.7	45.3	14.3
Ours PT (10%)	35.0	27.9	114.1	21.3	28.8	22.1	61.7	16.8
Ours FT (10%)	32.7	27.5	109.0	21.1	26.9	21.6	52.1	14.5
Ours FT (100%)	35.6	28.1	116.5	21.5	30.1	24.5	72.7	18.2

Table 4. Image caption performance under different tuning settings. B@4, M, C, S stand for BLEU-4, METEOR, CIDEr, and SPICE scores, respectively.

Task	MSVD				
	B@4	M	R	C	S
ORG-TRL [75]	54.3	36.4	73.9	95.2	-
Ours w/o Tuning	20.3	25.8	52.1	45.7	6.5
Ours PT (1%)	54.8	38.9	74.7	104.8	6.6
Ours FT (1%)	47.3	35.8	66.2	80.1	6.2
Ours PT (10%)	57.2	39.1	75.6	112.1	6.8
Ours FT (10%)	56.7	38.7	70.0	88.2	6.7
Ours FT (100%)	61.5	42.3	79.0	131.0	7.7

Table 5. Video caption (novel task) performance under different tuning settings. Note that this task did not appear in our pre-training. The only task related to video modality in our pre-training is video classification.

of zero-shot inference on novel tasks, and can reach the performance close to SOTA results by prompt tuning with only a small amount of downstream data. The performance can be further improved to be on par with or superior to SOTA results by full-data fine-tuning.

Limitations. Our method is currently only applicable when the target set is discrete, such as classification and retrieval. Whether our model can be extended to regression tasks is still questionable. Future work may explore the unified perception model of both discrete and continuous target sets.

Task	Text Retrieval MSVD			Video Retrieval MSVD		
	R@1	R@5	R10	R@1	R@5	R@10
CLIP4clip [48]	56.6	79.7	84.3	46.2	76.1	84.6
CLIP2video [18]	58.7	85.6	91.6	47.0	76.8	85.9
Ours w/o Tuning	42.7	69.1	79.6	34.6	64.5	75.4
Ours PT (1%)	61.2	83.7	89.0	42.6	73.3	82.5
Ours FT (1%)	49.6	75.8	83.7	37.5	68.2	79.3
Ours PT (10%))	61.3	84.8	90.9	43.1	74.2	83.4
Ours FT (10%))	59.1	81.9	87.4	41.7	71.6	81.3
Ours FT (100%)	61.5	83.5	90.2	45.4	75.8	85.0

Table 6. Video-text retrieval (novel task) performance under different tuning settings. Note that this task did not appear in our pre-training.

Task	VQA v2 test-dev		
	Yes/No	Numbers	Others
Unified VLP [78]	87.2	52.1	60.3
Ours w/o Tuning	0.9	3.0	25.5
Ours PT (0.1%)	63.0	31.8	49.6
Ours FT (0.1%)	63.0	30.6	49.1
Ours PT (1%)	70.8	41.3	57.7
Ours FT (1%)	71.0	42.4	57.5
Ours FT (100%)	84.8	47.4	61.8

Table 7. Visual question answering (novel task) performance under different tuning settings. Note that this task did not appear in our pre-training.

Potential Negative Societal Impact. This work may share the common negative impacts of large-scale training, which may consume lots of electricity and result in increased carbon emissions. This method also learns from a large number of datasets that may contain data biases. Future work may seek for more efficient and unbiased training.

Task	Text Retrieval						Image Retrieval					
	Flickr30k			COCO Caption			Flickr30k			COCO Caption		
	R@1	R@5	R10	R@1	R@5	R10	R@1	R@5	R10	R@1	R@5	R10
ImageBERT [53] w/o Tuning	70.7	90.2	94.0	44.0	71.2	80.4	54.3	79.6	87.5	32.3	59.0	70.2
UNITER-B [14] w/o Tuning	80.7	95.7	98.0	-	-	-	66.2	88.4	92.9	-	-	-
ViLT [28] w/o Tuning	73.2	93.6	96.5	56.5	82.6	89.6	55.0	82.5	89.8	40.4	70.0	81.1
Unicoder-VL [23]	86.2	96.3	99.0	62.3	87.1	92.8	71.5	91.2	95.2	48.4	76.7	85.9
UNITER-B	85.9	97.1	98.8	64.4	87.4	93.1	72.5	92.4	96.1	50.3	78.5	87.2
ViLT	83.5	96.7	98.6	61.5	86.3	92.7	64.4	88.7	93.8	42.7	72.9	83.1
Ours w/o Tuning	74.8	94.8	98.2	57.7	85.6	92.3	65.8	88.8	93.6	46.3	75.0	84.0
Ours PT (1%)	84.4	97.8	99.2	61.4	86.7	93.2	71.1	91.6	95.1	47.0	75.3	84.3
Ours FT (1%)	78.4	95.7	97.8	60.2	85.1	90.6	61.0	85.7	91.0	43.6	70.9	80.5
Ours PT (10%)	86.4	98.2	99.5	61.6	87.0	93.2	72.5	92.3	95.7	47.2	75.4	84.3
Ours FT (10%)	84.9	97.4	98.3	60.9	85.5	92.1	67.9	89.4	92.9	45.6	73.4	82.6
Ours FT (100%)	87.9	98.2	99.1	64.7	87.8	93.7	74.9	93.5	96.0	48.3	75.9	84.5

表 3. 不同调优设置下的图像-文本检索性能。PT 表示提示调优，FT 表示微调。标注了用于调优的数据比例。

Task	COCO Caption				Flickr30k			
	B@4	M	C	S	B@4	M	C	S
Unified VLP [78]	36.5	28.4	116.9	21.2	30.1	23.0	67.4	17.0
Ours w/o Tuning	33.6	27.0	109.8	20.3	17.0	16.2	41.2	11.2
Ours PT (1%)	34.3	27.2	109.6	21.2	28.1	21.6	59.1	15.6
Ours FT (1%)	28.0	26.8	100.1	20.2	18.9	19.7	45.3	14.3
Ours PT (10%)	35.0	27.9	114.1	21.3	28.8	22.1	61.7	16.8
Ours FT (10%)	32.7	27.5	109.0	21.1	26.9	21.6	52.1	14.5
Ours FT (100%)	35.6	28.1	116.5	21.5	30.1	24.5	72.7	18.2

表4. 不同调优设置下的图像描述性能。B@4、M、C、S分别表示BLEU-4、METEOR、CIDEr和SPICE分数。

Task	MSVD				
	B@4	M	R	C	S
ORG-TRL [75]	54.3	36.4	73.9	95.2	-
Ours w/o Tuning	20.3	25.8	52.1	45.7	6.5
Ours PT (1%)	54.8	38.9	74.7	104.8	6.6
Ours FT (1%)	47.3	35.8	66.2	80.1	6.2
Ours PT (10%)	57.2	39.1	75.6	112.1	6.8
Ours FT (10%)	56.7	38.7	70.0	88.2	6.7
Ours FT (100%)	61.5	42.3	79.0	131.0	7.7

表 5. 不同调优设置下的视频字幕生成（新任务）性能。请注意，该任务未出现在我们的预训练中。我们的预训练中唯一与视频模态相关的任务是视频分类。

在新任务上进行零样本推理，并且仅使用少量下游数据进行提示调优，就能达到接近 SOTA 结果的性能。通过全数据微调，性能还可以进一步提升，达到与 SOTA 结果相当或优于 SOTA 结果的水平。

局限性。我们的方法目前仅适用于目标集是离散的情况，例如分类和检索。我们的模型能否扩展到回归任务仍有待商榷。未来的工作可以探索统一的 per-

Task	Text Retrieval MSVD			Video Retrieval MSVD		
	R@1	R@5	R10	R@1	R@5	R@10
CLIP4clip [48]	56.6	79.7	84.3	46.2	76.1	84.6
CLIP2video [18]	58.7	85.6	91.6	47.0	76.8	85.9
Ours w/o Tuning	42.7	69.1	79.6	34.6	64.5	75.4
Ours PT (1%)	61.2	83.7	89.0	42.6	73.3	82.5
Ours FT (1%)	49.6	75.8	83.7	37.5	68.2	79.3
Ours PT (10%))	61.3	84.8	90.9	43.1	74.2	83.4
Ours FT (10%))	59.1	81.9	87.4	41.7	71.6	81.3
Ours FT (100%)	61.5	83.5	90.2	45.4	75.8	85.0

表6. 不同调优设置下的视频-文本检索（新任务）性能。请注意，该任务未出现在我们的预训练中。

Task	VQA v2 test-dev		
	Yes/No	Numbers	Others
Unified VLP [78]	87.2	52.1	60.3
Ours w/o Tuning	0.9	3.0	25.5
Ours PT (0.1%)	63.0	31.8	49.6
Ours FT (0.1%)	63.0	30.6	49.1
Ours PT (1%)	70.8	41.3	57.7
Ours FT (1%)	71.0	42.4	57.5
Ours FT (100%)	84.8	47.4	61.8

表 7. 不同调优设置下的视觉问答（新任务）性能。请注意，该任务并未出现在我们的预训练中。

离散和连续目标集的感知模型。

潜在的负面社会影响。此项工作可能具有大规模训练常见的负面影响，即可能消耗大量电力并导致碳排放增加。该方法还从大量数据集中学习，而这些数据集可能包含数据偏差。未来的工作可以寻求更高效且无偏的训练方法。

Task	GLUE					
	MNLI (Acc)	QNLI (Acc)	QQP (F1)	RTE (Acc)	SST-2 (Acc)	MRPC (F1)
PLM [3] w/o tuning	49.4	50.7	46.6	53.8	70.6	44.2
$\text{BERT}_{\text{BASE}}$ [70]	84.6	92.7	71.2	66.4	93.5	88.9
Ours w/o Tuning	49.6	51.0	53.6	55.6	70.6	76.1
Ours PT (1%)	60.1	76.0	70.2	56.3	80.9	80.3
Ours FT (1%)	47.3	60.6	68.9	49.1	69.7	72.3
Ours PT (10%)	68.5	83.2	77.0	58.2	83.4	83.2
Ours FT (10%)	60.5	71.5	71.4	50.5	79.1	80.6
Ours FT (100%)	81.7	89.9	87.1	64.3	90.2	86.6

Table 8. Natural language understanding (novel task) performance under different tuning settings. Note that this task did not appear in our pre-training.

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References

[1] Hassan Akbari, Li Yuan, Rui Qian, Wei-Hong Chuang, Shih-Fu Chang, Yin Cui, and Boqing Gong. Vatt: Transformers for multimodal self-supervised learning from raw video, audio and text. *arXiv preprint arXiv:2104.11178*, 2021. 2, 3

[2] Chris Alberti, Jeffrey Ling, Michael Collins, and David Reitter. Fusion of detected objects in text for visual question answering. *arXiv preprint arXiv:1908.05054*, 2019. 3

[3] Anonymous. Are BERT families zero-shot learners? a study on their potential and limitations. In *Submitted to ICLR*, 2022. under review. 7, 9

[4] Anurag Arnab, Mostafa Dehghani, Georg Heigold, Chen Sun, Mario Lučić, and Cordelia Schmid. Vivit: A video vision transformer. *arXiv preprint arXiv:2103.15691*, 2021. 2

[5] Gedas Bertasius, Heng Wang, and Lorenzo Torresani. Is space-time attention all you need for video understanding? In *ICML*, July 2021. 2

[6] Gedas Bertasius, Heng Wang, and Lorenzo Torresani. Is space-time attention all you need for video understanding? *arXiv preprint arXiv:2102.05095*, 2021. 4, 7, 12, 14

[7] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, T. J. Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeff Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language models are few-shot learners. *arXiv preprint arXiv:2005.14165*, 2020. 2, 3, 5, 6

[8] Soravit Changpinyo, Piyush Sharma, Nan Ding, and Radu Soricut. Conceptual 12M: Pushing web-scale image-text pre-training to recognize long-tail visual concepts. In *CVPR*, 2021. 3, 6, 14

[9] Chun-Fu Chen, Quanfu Fan, and Rameswar Panda. Crossvit: Cross-attention multi-scale vision transformer for image classification. *arXiv preprint arXiv:2103.14899*, 2021. 2

[10] David Chen and William B Dolan. Collecting highly parallel data for paraphrase evaluation. In *ACL*, pages 190–200, 2011. 6, 7

[11] Xinlei Chen, Hao Fang, Tsung-Yi Lin, Ramakrishna Vedantam, Saurabh Gupta, Piotr Dollár, and C. Lawrence Zitnick. Microsoft coco captions: Data collection and evaluation server. *arXiv preprint arXiv:1504.00325*, 2015. 6, 14

[12] Xinlei Chen, Hao Fang, Tsung-Yi Lin, Ramakrishna Vedantam, Saurabh Gupta, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco captions: Data collection and evaluation server. *arXiv preprint arXiv:1504.00325*, 2015. 6

[13] Xinyang Chen, Sinan Wang, Bo Fu, Mingsheng Long, and Jianmin Wang. Catastrophic forgetting meets negative transfer: Batch spectral shrinkage for safe transfer learning. In *NeurIPS*, 2019. 2

[14] Yen-Chun Chen, Linjie Li, Licheng Yu, Ahmed El Kholy, Faisal Ahmed, Zhe Gan, Yu Cheng, and Jingjing Liu. Uniter: Universal image-text representation learning. In *ECCV*, pages 104–120. Springer, 2020. 2, 3, 8

[15] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *CVPR*, pages 248–255. Ieee, 2009. 3, 6, 14

[16] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018. 2, 3, 5, 6

[17] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020. 2, 4, 12

[18] Han Fang, Pengfei Xiong, Luhui Xu, and Yu Chen. Clip2video: Mastering video-text retrieval via image clip. *arXiv preprint arXiv:2106.11097*, 2021. 8

[19] Tianyu Gao, Adam Fisch, and Danqi Chen. Making pre-trained language models better few-shot learners. *arXiv preprint arXiv:2012.15723*, 2020. 7

[20] Tianyu Gao, Adam Fisch, and Danqi Chen. Making pre-trained language models better few-shot learners. In *ACL*, 2021. 3, 6

[21] Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. Making the V in VQA matter: Elevating the role of image understanding in Visual Question Answering. In *CVPR*, 2017. 6, 7

[22] Ronghang Hu and Amanpreet Singh. Unit: Multimodal multitask learning with a unified transformer. *arXiv preprint arXiv:2102.10772*, 2021. 2, 3

[23] Haoyang Huang, Yaobo Liang, Nan Duan, Ming Gong, Linjun Shou, Daxin Jiang, and Ming Zhou. Unicoder: A universal language encoder by pre-training with multiple cross-lingual tasks. *arXiv preprint arXiv:1909.00964*, 2019. 8

[24] Zhicheng Huang, Zhaoyang Zeng, Bei Liu, Dongmei Fu, and Jianlong Fu. Pixel-bert: Aligning image pixels with text by deep multi-modal transformers. *arXiv preprint arXiv:2004.00849*, 2020. 3

[25] Zhengbao Jiang, Frank F. Xu, J. Araki, and Graham Neubig. How can we know what language models know? *TACL*, 8:423–438, 2020. 3

Task	GLUE					
	MNLI (Acc)	QNLI (Acc)	QQP (F1)	RTE (Acc)	SST-2 (Acc)	MRPC (F1)
PLM [3] $_{\text{w/o tuning}}$	49.4	50.7	46.6	53.8	70.6	44.2
$\text{BERT}_{\text{BASE}}$ [70]	84.6	92.7	71.2	66.4	93.5	88.9
Ours $_{\text{w/o Tuning}}$	49.6	51.0	53.6	55.6	70.6	76.1
Ours $_{\text{PT (1\%)}}$	60.1	76.0	70.2	56.3	80.9	80.3
Ours $_{\text{FT (1\%)}}$	47.3	60.6	68.9	49.1	69.7	72.3
Ours $_{\text{PT (10\%)}}$	68.5	83.2	77.0	58.2	83.4	83.2
Ours $_{\text{FT (10\%)}}$	60.5	71.5	71.4	50.5	79.1	80.6
Ours $_{\text{FT (100\%)}}$	81.7	89.9	87.1	64.3	90.2	86.6

表 8. 不同调优设置下的自然语言理解（新任务）性能。请注意，该任务未出现在我们的预训练中。

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参考文献

[1] Hassan Akbari, Li Yuan, Rui Qian, Wei-Hong Chuang, Shih-Fu Chang, Yin Cui, 和 Boqing Gong。Vatt：用于从原始视频、音频和文本进行多模态自监督学习的 Transformer。arXiv 预印本 arXiv:2104.11178，2021。2, 3 [2] Chris Alberti, Jeffrey Ling, Michael Collins, 和 David Reitter。视觉问答中检测到的对象在文本中的融合。arXiv 预印本 arXiv:1908.05054，2019。3 [3] Anonymous。BERT 家族是零样本学习器吗？关于其潜力与局限性的研究。提交至 ICLR，2022。审稿中。7, 9 [4] Anurag Arnab, Mostafa Dehghani, Georg Heigold, Chen Sun, Mario Lučić, 和 Cordelia Schmid。Vivit：视频视觉 Transformer。arXiv 预印本 arXiv:2103.15691，2021。2 [5] Gedas Bertasius, Heng Wang, 和 Lorenzo Torresani。时空注意力就是视频理解所需的一切吗？发表于 ICML，2021 年 7 月。2 [6] Gedas Bertasius, Heng Wang, 和 Lorenzo Torresani。时空注意力就是视频理解所需的一切吗？arXiv 预印本 arXiv:2102.05095，2021。4, 7, 12, 14 [7] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, T. J. Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeff Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, 和 Dario Amodei。语言模型是少样本学习器。arXiv 预印本 arXiv:2005.14165，2020。2, 3, 5, 6 [8] Soravit Changpinyo, Piyush Sharma, Nan Ding, 和 Radu Soricut。Conceptual 12M：将网络规模的图文预训练推进到识别长尾视觉概念。发表于 CVPR，2021。3, 6, 14 [9] Chun-Fu Chen, Quanfu Fan, 和 Rameswar Panda。Crossvit：用于图像分类的交叉注意力多尺度视觉 Transformer。arXiv 预印本 arXiv:2103.14899，2021。2

[10] David Chen 和 William B Dolan。收集高度并行的数据用于释义评估。发表于 ACL，第 190–200 页，2011。6, 7 [11] Xinlei Chen、Hao Fang、Tsung-Yi Lin、Ramakrishna Vedantam、Saurabh Gupta、Piotr Dollár 和 C. Lawrence Zit-nick。Microsoft COCO captions：数据收集与评估服务器。arXiv 预印本 arXiv:1504.00325，2015。6, 14 [12] Xinlei Chen、Hao Fang、Tsung-Yi Lin、Ramakrishna Vedan-tam、Saurabh Gupta、Piotr Dollár 和 C Lawrence Zitnick。Microsoft COCO captions：数据收集与评估服务器。arXiv 预印本 arXiv:1504.00325，2015。6 [13] Xinyang Chen、Sinan Wang、Bo Fu、Mingsheng Long 和 Jianmin Wang。灾难性遗忘遇上负迁移：用于安全迁移学习的批谱收缩。发表于 NeurIPS，2019。2 [14] Yen-Chun Chen、Linjie Li、Licheng Yu、Ahmed El Kholy、Faisal Ahmed、Zhe Gan、Yu Cheng 和 Jingjing Liu。Uniter：通用图像-文本表征学习。发表于 ECCV，第 104–120 页。Springer，2020。2, 3, 8 [15] Jia Deng、Wei Dong、Richard Socher、Li-Jia Li、Kai Li 和 Li Fei-Fei。Imagenet：一个大规模层次化图像数据库。发表于 CVPR，第 248–255 页。IEEE，2009。3, 6, 14 [16] Jacob Devlin、Ming-Wei Chang、Kenton Lee 和 Kristina Toutanova。Bert：用于语言理解的深度双向 Transformer 预训练。arXiv 预印本 arXiv:1810.04805，2018。2, 3, 5, 6 [17] Alexey Dosovitskiy、Lucas Beyer、Alexander Kolesnikov、Dirk Weissenborn、Xiaohua Zhai、Thomas Unterthiner、Mostafa Dehghani、Matthias Minderer、Georg Heigold、Syl-vain Gelly 等。一张图像胜过 16x16 个词：用于大规模图像识别的 Transformer。arXiv 预印本 arXiv:2010.11929，2020。2, 4, 12 [18] Han Fang、Pengfei Xiong、Luhui Xu 和 Yu Chen。Clip2video：通过图像 CLIP 掌握视频-文本检索。arXiv 预印本 arXiv:2106.11097，2021。8 [19] Tianyu Gao、Adam Fisch 和 Danqi Chen。让预训练语言模型成为更好的少样本学习者。arXiv 预印本 arXiv:2012.15723，2020。7 [20] Tianyu Gao、Adam Fisch 和 Danqi Chen。让预训练语言模型成为更好的少样本学习者。发表于 ACL，2021。3, 6 [21] Yash Goyal、Tejas Khot、Douglas Summers-Stay、Dhruv Ba-tra 和 Devi Parikh。让 VQA 中的 V 变得重要：提升图像理解在视觉问答中的作用。发表于 CVPR，2017。6, 7 [22] Ronghang Hu 和 Amanpreet Singh。Unit：使用统一 Transformer 的多模态多任务学习。arXiv 预印本 arXiv:2102.10772，2021。2, 3 [23] Haoyang Huang、Yaobo Liang、Nan Duan、Ming Gong、Lin-jun Shou、Daxin Jiang 和 Ming Zhou。Unicoder：通过多个跨语言任务预训练得到的通用语言编码器。arXiv 预印本 arXiv:1909.00964，2019。8 [24] Zhicheng Huang、Zhaoyang Zeng、Bei Liu、Dongmei Fu 和 Jianlong Fu。Pixel-bert：通过深度多模态 Transformer 将图像像素与文本对齐。arXiv 预印本 arXiv:2004.00849，2020。3 [25] Zhengbao Jiang、Frank F. Xu、J. Araki 和 Graham Neubig。我们如何知道语言模型知道什么？TACL，8:423–438，2020。3

[26] Sebastian Kalkowski, Christian Schulze, Andreas Dengel, and Damian Borth. Real-time analysis and visualization of the yfcc100m dataset. In *Proceedings of the 2015 workshop on community-organized multimodal mining: opportunities for novel solutions*, pages 25–30, 2015. 3, 6, 14

[27] Will Kay, Joao Carreira, Karen Simonyan, Brian Zhang, Chloe Hillier, Sudheendra Vijayanarasimhan, Fabio Viola, Tim Green, Trevor Back, Paul Natsev, et al. The kinetics human action video dataset. *arXiv preprint arXiv:1705.06950*, 2017. 3, 6, 14

[28] Wonjae Kim, Bokyung Son, and Ildoo Kim. Vilt: Vision-and-language transformer without convolution or region supervision. *arXiv preprint arXiv:2102.03334*, 2021. 2, 3, 8

[29] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014. 6

[30] Ranjay Krishna, Yuke Zhu, Oliver Groth, Justin Johnson, Kenji Hata, Joshua Kravitz, Stephanie Chen, Yannis Kalantidis, Li-Jia Li, David A Shamma, et al. Visual genome: Connecting language and vision using crowdsourced dense image annotations. *IJCV*, 123(1):32–73, 2017. 3, 6, 14

[31] Zhenzhong Lan, Mingda Chen, Sebastian Goodman, Kevin Gimpel, Piyush Sharma, and Radu Soricut. Albert: A lite bert for self-supervised learning of language representations. *arXiv preprint arXiv:1909.11942*, 2019. 2, 3

[32] Gustav Larsson, Michael Maire, and Gregory Shakhnarovich. Fractalnet: Ultra-deep neural networks without residuals. *arXiv preprint arXiv:1605.07648*, 2017. 6

[33] Brian Lester, Rami Al-Rfou, and Noah Constant. The power of scale for parameter-efficient prompt tuning. In *EMNLP*, 2021. 3

[34] Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Ves Stoyanov, and Luke Zettlemoyer. Bart: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension. *arXiv preprint arXiv:1910.13461*, 2019. 2, 3

[35] Patrick Lewis, Yuxiang Wu, Linqing Liu, Pasquale Minervini, Heinrich Küttler, Aleksandra Piktus, Pontus Stenetorp, and Sebastian Riedel. Paq: 65 million probably-asked questions and what you can do with them. *arXiv preprint arXiv:2102.07033*, 2021. 6, 14

[36] Gen Li, Nan Duan, Yuejian Fang, Daxin Jiang, and Ming Zhou. Unicoder-vl: A universal encoder for vision and language by cross-modal pre-training. In *AAAI*, 2020. 2

[37] Liunian Harold Li, Mark Yatskar, Da Yin, Cho-Jui Hsieh, and Kai-Wei Chang. Visualbert: A simple and performant baseline for vision and language. *arXiv preprint arXiv:1908.03557*, 2019. 2, 3

[38] Xiang Lisa Li and Percy Liang. Prefix-tuning: Optimizing continuous prompts for generation, 2021. 3

[39] Zewen Li, Fan Liu, Wenjie Yang, Shouheng Peng, and Jun Zhou. A survey of convolutional neural networks: Analysis, applications, and prospects. *IEEE Transactions on Neural Networks and Learning Systems*, pages 1–21, 2021. 1, 2

[40] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco: Common objects in context. In *ECCV*, 2014. 3

[41] Pengfei Liu, Weizhe Yuan, Jinlan Fu, Zhengbao Jiang, Hiroaki Hayashi, and Graham Neubig. Pre-train, prompt, and predict: A systematic survey of prompting methods in natural language processing. *arXiv preprint arXiv:2107.13586*, 2021. 2, 3, 6

[42] Xiao Liu, Kaixuan Ji, Yicheng Fu, Zhengxiao Du, Zhilin Yang, and Jie Tang. P-tuning v2: Prompt tuning can be comparable to fine-tuning universally across scales and tasks. *arXiv preprint arXiv:2110.07602*, 2021. 3, 6

[43] Xiao Liu, Yanan Zheng, Zhengxiao Du, Ming Ding, Yujie Qian, Zhilin Yang, and Jie Tang. Gpt understands, too. *arXiv preprint arXiv:2103.10385*, 2021. 3

[44] Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692*, 2019. 2, 3

[45] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. *ICCV*, 2021. 2

[46] Jiasen Lu, Dhruv Batra, Devi Parikh, and Stefan Lee. Vilbert: Pretraining task-agnostic visiolinguistic representations for vision-and-language tasks. In *NeurIPS*, 2019. 2

[47] Jiasen Lu, Dhruv Batra, Devi Parikh, and Stefan Lee. Vilbert: Pretraining task-agnostic visiolinguistic representations for vision-and-language tasks. *arXiv preprint arXiv:1908.02265*, 2019. 3

[48] Huaishao Luo, Lei Ji, Ming Zhong, Yang Chen, Wen Lei, Nan Duan, and Tianrui Li. Clip4clip: An empirical study of clip for end to end video clip retrieval. *arXiv preprint arXiv:2104.08860*, 2021. 8

[49] Mathew Monfort, Alex Andonian, Bolei Zhou, Kandan Ramakrishnan, Sarah Adel Bargal, Tom Yan, Lisa Brown, Quanfu Fan, Dan Gutfreund, Carl Vondrick, et al. Moments in time dataset: one million videos for event understanding. *TPAMI*, 42(2):502–508, 2019. 3, 6, 14

[50] Vicente Ordonez, Girish Kulkarni, and Tamara Berg. Im2text: Describing images using 1 million captioned photographs. *NeurIPS*, 24:1143–1151, 2011. 3, 6, 14

[51] Gao Peng, Geng Shijie, Zhang Renrui, Ma Teli, Fang Rongyao, Zhang Yongfeng, Li Hongsheng, and Qiao Yu. Clip-adapter: Better vision-language models with feature adapters. *arXiv preprint arXiv:2110.04544*, 2021. 3

[52] Bryan A Plummer, Liwei Wang, Chris M Cervantes, Juan C Caicedo, Julia Hockenmaier, and Svetlana Lazebnik. Flickr30k entities: Collecting region-to-phrase correspondences for richer image-to-sentence models. In *ICCV*, pages 2641–2649, 2015. 6

[53] Di Qi, Lin Su, Jianwei Song, Edward Cui, Taroon Bharti, and Arun Sacheti. Imagebert: Cross-modal pre-training with large-scale weak-supervised image-text data. *arXiv preprint arXiv:2001.07966*, 2020. 2, 8

[54] Xipeng Qiu, Tianxiang Sun, Yige Xu, Yunfan Shao, Ning Dai, and Xuanjing Huang. Pre-trained models for natural language processing: A survey. *Science China Technological Sciences*, pages 1–26, 2020. 3

[55] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya

[26] Sebastian Kalkowski、Christian Schulze、Andreas Dengel 和 Damian Borth。yfcc100m 数据集的实时分析与可视化。载于《2015 年社区组织的多模态挖掘研讨会论文集：新颖解决方案的机会》，第 25–30 页，2015。3, 6, 14 [27] Will Kay、Joao Carreira、Karen Simonyan、Brian Zhang、Chloe Hillier、Sudheendra Vijayanarasimhan、Fabio Viola、Tim Green、Trevor Back、Paul Natsev 等。Kinetics 人体动作视频数据集。arXiv 预印本 arXiv:1705.06950，2017。3, 6, 14 [28] Wonjae Kim、Bokyung Son 和 Ildoo Kim。Vilt：无需卷积或区域监督的视觉-语言 Transformer。arXiv 预印本 arXiv:2102.03334，2021。2, 3, 8 [29] Diederik P Kingma 和 Jimmy Ba。Adam：一种随机优化方法。arXiv 预印本 arXiv:1412.6980，2014。6 [30] Ranjay Krishna、Yuke Zhu、Oliver Groth、Justin Johnson、Kenji Hata、Joshua Kravitz、Stephanie Chen、Yannis Kalantidis、Li-Jia Li、David A Shamma 等。Visual Genome：使用众包密集图像标注连接语言与视觉。IJCV，123(1):32–73，2017。3, 6, 14 [31] Zhenzhong Lan、Mingda Chen、Sebastian Goodman、Kevin Gimpel、Piyush Sharma 和 Radu Soricut。Albert：用于语言表示自监督学习的轻量级 BERT。arXiv 预印本 arXiv:1909.11942，2019。2, 3 [32] Gustav Larsson、Michael Maire 和 Gregory Shakhnarovich。Fractalnet：无残差的超深神经网络。arXiv 预印本 arXiv:1605.07648，2017。6 [33] Brian Lester、Rami Al-Rfou 和 Noah Constant。面向参数高效提示调优的规模力量。载于 EMNLP，2021。3 [34] Mike Lewis、Yinhan Liu、Naman Goyal、Marjan Ghazvininejad、Abdelrahman Mohamed、Omer Levy、Ves Stoyanov 和 Luke Zettlemoyer。Bart：用于自然语言生成、翻译和理解的去噪序列到序列预训练。arXiv 预印本 arXiv:1910.13461，2019。2, 3 [35] Patrick Lewis、Yuxiang Wu、Linqing Liu、Pasquale Minervini、Heinrich Küttler、Aleksandra Piktus、Pontus Stenetorp 和 Sebastian Riedel。Paq：6500 万个可能被问到的问题以及你可以用它们做什么。arXiv 预印本 arXiv:2102.07033，2021。6, 14 [36] Gen Li、Nan Duan、Yuejian Fang、Daxin Jiang 和 Ming Zhou。Unicoder-vl：通过跨模态预训练实现的视觉与语言通用编码器。载于 AAAI，2020。2 [37] Liunian Harold Li、Mark Yatskar、Da Yin、Cho-Jui Hsieh 和 Kai-Wei Chang。Visualbert：一个简单且高性能的视觉与语言基线。arXiv 预印本 arXiv:1908.03557，2019。2, 3 [38] Xiang Lisa Li 和 Percy Liang。前缀调优：为生成优化连续提示，2021。3 [39] Zewen Li、Fan Liu、Wenjie Yang、Shouheng Peng 和 Jun Zhou。卷积神经网络综述：分析、应用与前景。IEEE Transactions on Neural Networks and Learning Systems，第 1–21 页，2021。1, 2 [40] Tsung-Yi Lin、Michael Maire、Serge Belongie、James Hays、Pietro Perona、Deva Ramanan、Piotr Dollár 和 C Lawrence Zitnick。Microsoft coco：上下文中的常见物体。载于 ECCV，2014。3 [41] Pengfei Liu、Weizhe Yuan、Jinlan Fu、Zhengbao Jiang、Hiroaki Hayashi 和 Graham Neubig。预训练、提示与预测：自然语言处理中提示方法的系统性综述。arXiv 预印本 arXiv:2107.13586，2021。2, 3, 6 [42] Xiao Liu、Kaixuan Ji、Yicheng Fu、Zhengxiao Du、Zhilin Yang 和 Jie Tang。P-tuning v2：提示调优在各种规模和任务上都可以普遍媲美微调。arXiv 预印本 arXiv:2110.07602，2021。3, 6 [43] Xiao Liu、Yanan Zheng、Zhengxiao Du、Ming Ding、Yujie Qian、Zhilin Yang 和 Jie Tang。GPT 也能理解。arXiv 预印本 arXiv:2103.10385，2021。3 [44] Yinhan Liu、Myle Ott、Naman Goyal、Jingfei Du、Mandar Joshi、Danqi Chen、Omer Levy、Mike Lewis、Luke Zettlemoyer 和 Veselin Stoyanov。RoBERTa：一种鲁棒优化的 BERT 预训练方法。arXiv 预印本 arXiv:1907.11692，2019。2, 3 [45] Ze Liu、Yutong Lin、Yue Cao、Han Hu、Yixuan Wei、Zheng Zhang、Stephen Lin 和 Baining Guo。Swin Transformer：使用移位窗口的层次化视觉 Transformer。ICCV，2021。2 [46] Jiasen Lu、Dhruv Batra、Devi Parikh 和 Stefan Lee。ViLBERT：面向视觉与语言任务的任务无关视觉语言表示预训练。见 NeurIPS，2019。2 [47] Jiasen Lu、Dhruv Batra、Devi Parikh 和 Stefan Lee。ViLBERT：面向视觉与语言任务的任务无关视觉语言表示预训练。arXiv 预印本 arXiv:1908.02265，2019。3 [48] Huaishao Luo、Lei Ji、Ming Zhong、Yang Chen、Wen Lei、Nan Duan 和 Tianrui Li。Clip4clip：用于端到端视频片段检索的 CLIP 实证研究。arXiv 预印本 arXiv:2104.08860，2021。8 [49] Mathew Monfort、Alex Andonian、Bolei Zhou、Kandan Ramakrishnan、Sarah Adel Bargal、Tom Yan、Lisa Brown、Quanfu Fan、Dan Gutfreund、Carl Vondrick 等。Moments in Time 数据集：用于事件理解的一百万个视频。TPAMI，42(2):502–508，2019。3, 6, 14 [50] Vicente Ordonez、Girish Kulkarni 和 Tamara Berg。Im2text：使用 100 万张带说明文字的照片描述图像。NeurIPS，24:1143–1151，2011。3, 6, 14 [51] Gao Peng、Geng Shijie、Zhang Renrui、Ma Teli、Fang Rongyao、Zhang Yongfeng、Li Hongsheng 和 Qiao Yu。CLIP-adapter：使用特征适配器改进视觉-语言模型。arXiv 预印本 arXiv:2110.04544，2021。3 [52] Bryan A Plummer、Liwei Wang、Chris M Cervantes、Juan C Caicedo、Julia Hockenmaier 和 Svetlana Lazebnik。Flickr30k 实体：为更丰富的图像到句子模型收集区域到短语对应关系。见 ICCV，第 2641–2649 页，2015。6 [53] Di Qi、Lin Su、Jianwei Song、Edward Cui、Taroon Bharti 和 Arun Sacheti。ImageBERT：使用大规模弱监督图像-文本数据进行跨模态预训练。arXiv 预印本 arXiv:2001.07966，2020。2, 8 [54] Xipeng Qiu、Tianxiang Sun、Yige Xu、Yunfan Shao、Ning Dai 和 Xuanjing Huang。自然语言处理的预训练模型：综述。Science China Technological Sciences，第 1–26 页，2020。3 [55] Alec Radford、Jong Wook Kim、Chris Hallacy、Aditya

Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. *arXiv preprint arXiv:2103.00020*, 2021. 3

[56] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *ICML*, 2021. 2, 3

[57] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. Faster r-cnn: Towards real-time object detection with region proposal networks. *NeurIPS*, 28:91–99, 2015. 3

[58] Zhang Renrui, Fang Rongyao, Gao Peng, Zhang Wei, Li Kunchang, Dai Jifeng, Qiao Yu, and Li Hongsheng. Tip-adapter: Training-free clip-adapter for better vision-language modeling. *arXiv preprint arXiv:2111.03930*, 2021. 3

[59] Rico Sennrich, Barry Haddow, and Alexandra Birch. Neural machine translation of rare words with subword units. In *ACL*, 2016. 4, 12

[60] Piyush Sharma, Nan Ding, Sebastian Goodman, and Radu Soricut. Conceptual captions: A cleaned, hypernymed, image alt-text dataset for automatic image captioning. In *ACL*, pages 2556–2565, 2018. 3, 6, 14

[61] Taylor Shin, Yasaman Razeghi, Robert L. Logan IV, Eric Wallace, and Sameer Singh. AutoPrompt: Eliciting knowledge from language models with automatically generated prompts. In *EMNLP*, 2020. 3

[62] Weijie Su, Xizhou Zhu, Yue Cao, Bin Li, Lewei Lu, Furu Wei, and Jifeng Dai. Vl-bert: Pre-training of generic visual-linguistic representations. *arXiv preprint arXiv:1908.08530*, 2019. 2, 3

[63] Chen Sun, Fabien Baradel, Kevin Murphy, and Cordelia Schmid. Learning video representations using contrastive bidirectional transformer. *arXiv preprint arXiv:1906.05743*, 2019. 3

[64] Chen Sun, Austin Myers, Carl Vondrick, Kevin Murphy, and Cordelia Schmid. Videobert: A joint model for video and language representation learning. In *CVPR*, pages 7464–7473, 2019. 3

[65] Hao Tan and Mohit Bansal. Lxmert: Learning cross-modality encoder representations from transformers. *arXiv preprint arXiv:1908.07490*, 2019. 2, 3

[66] Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles, and Herve Jegou. Training data-efficient image transformers & distillation through attention. In *ICML*, volume 139, pages 10347–10357, July 2021. 2

[67] Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles, and Hervé Jégou. Training data-efficient image transformers & distillation through attention. In *ICML*, pages 10347–10357. PMLR, 2021. 7, 14

[68] Hugo Touvron, Matthieu Cord, Alexandre Sablayrolles, Gabriel Synnaeve, and Hervé Jégou. Going deeper with image transformers. *arXiv preprint arXiv:2103.17239*, 2021. 2

[69] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *NeurIPS*, pages 5998–6008, 2017. 1, 2

[70] Alex Wang, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel R Bowman. Glue: A multi-task benchmark and analysis platform for natural language understanding. *arXiv preprint arXiv:1804.07461*, 2018. 6, 7, 9

[71] Wenhai Wang, Enze Xie, Xiang Li, Deng-Ping Fan, Kaitao Song, Ding Liang, Tong Lu, Ping Luo, and Ling Shao. Pyramid vision transformer: A versatile backbone for dense prediction without convolutions. In *ICCV*, 2021. 2

[72] Han Xu, Zhang Zhengyan, Ding Ning, Gu Yuxian, Liu Xiao, Huo Yuqi, Qiu Jiezhong, Zhang Liang, Han Wentao, Huang Minlie, et al. Pre-trained models: Past, present and future. *arXiv preprint arXiv:2106.07139*, 2021. 3

[73] Li Yuan, Yunpeng Chen, Tao Wang, Weihao Yu, Yujun Shi, Zi-Hang Jiang, Francis E.H. Tay, Jiashi Feng, and Shuicheng Yan. Tokens-to-token vit: Training vision transformers from scratch on imagenet. In *ICCV*, pages 558–567, October 2021. 2

[74] Yanyi Zhang, Xinyu Li, Chunhui Liu, Bing Shuai, Yi Zhu, Biagio Brattoli, Hao Chen, Ivan Marsic, and Joseph Tighe. Vidtr: Video transformer without convolutions. In *ICCV*, pages 13577–13587, October 2021. 2

[75] Ziqi Zhang, Yaya Shi, Chunfeng Yuan, Bing Li, Peijin Wang, Weiming Hu, and Zheng-Jun Zha. Object relational graph with teacher-recommended learning for video captioning. In *CVPR*, pages 13278–13288, 2020. 8

[76] Daquan Zhou, Bingyi Kang, Xiaojie Jin, Linjie Yang, Xiaochen Lian, Qibin Hou, and Jiashi Feng. Deepvit: Towards deeper vision transformer. *arXiv preprint arXiv:2103.11886*, 2021. 2

[77] Kaiyang Zhou, Jingkang Yang, Chen Change Loy, and Ziwei Liu. Learning to prompt for vision-language models. *arXiv preprint arXiv:2109.01134*, 2021. 3

[78] Luowei Zhou, Hamid Palangi, Lei Zhang, Houdong Hu, Jason Corso, and Jianfeng Gao. Unified vision-language pre-training for image captioning and vqa. In *AAAI*, 2020. 2, 8

[79] Yukun Zhu, Ryan Kiros, Rich Zemel, Ruslan Salakhutdinov, Raquel Urtasun, Antonio Torralba, and Sanja Fidler. Aligning books and movies: Towards story-like visual explanations by watching movies and reading books. In *ICCV*, pages 19–27, 2015. 6, 14

Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark 等。通过自然语言监督学习可迁移的视觉模型。arXiv 预印本 arXiv:2103.00020，2021。3 [56] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger 和 Ilya Sutskever。通过自然语言监督学习可迁移的视觉模型。发表于 ICML，2021。2, 3 [57] Shaoqing Ren, Kaiming He, Ross Girshick 和 Jian Sun。Faster r-cnn：利用区域提议网络实现实时目标检测。NeurIPS，28:91–99，2015。3 [58] Zhang Renrui, Fang Rongyao, Gao Peng, Zhang Wei, Li Kunchang, Dai Jifeng, Qiao Yu 和 Li Hongsheng。Tip-adapter：用于更好视觉-语言建模的免训练 clip 适配器。arXiv 预印本 arXiv:2111.03930，2021。3 [59] Rico Sennrich, Barry Haddow 和 Alexandra Birch。使用子词单元进行稀有词的神经机器翻译。发表于 ACL，2016。4, 12 [60] Piyush Sharma, Nan Ding, Sebastian Goodman 和 Radu Soricut。Conceptual captions：一个经过清洗、上位词化的图像替代文本数据集，用于自动图像描述生成。发表于 ACL，页 2556–2565，2018。3, 6, 14 [61] Taylor Shin, Yasaman Razeghi, Robert L. Logan IV, Eric Wallace 和 Sameer Singh。AutoPrompt：使用自动生成的提示从语言模型中引出知识。发表于 EMNLP，2020。3 [62] Weijie Su, Xizhou Zhu, Yue Cao, Bin Li, Lewei Lu, Furu Wei 和 Jifeng Dai。Vl-bert：通用视觉-语言表示的预训练。arXiv 预印本 arXiv:1908.08530，2019。2, 3 [63] Chen Sun, Fabien Baradel, Kevin Murphy 和 Cordelia Schmid。使用对比式双向 Transformer 学习视频表示。arXiv 预印本 arXiv:1906.05743，2019。3 [64] Chen Sun, Austin Myers, Carl Vondrick, Kevin Murphy 和 Cordelia Schmid。Videobert：用于视频和语言表示学习的联合模型。发表于 CVPR，页 7464–7473，2019。3 [65] Hao Tan 和 Mohit Bansal。Lxmert：从 Transformer 学习跨模态编码器表示。arXiv 预印本 arXiv:1908.07490，2019。2, 3 [66] Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles 和 Herve Jegou。训练数据高效的图像 Transformer 与通过注意力进行蒸馏。发表于 ICML，第 139 卷，页 10347–10357，2021 年 7 月。2 [67] Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles 和 Hervé Jégou。训练数据高效的图像 Transformer 与通过注意力进行蒸馏。发表于 ICML，页 10347–10357。PMLR，2021。7, 14 [68] Hugo Touvron, Matthieu Cord, Alexandre Sablayrolles, Gabriel Synnaeve 和 Hervé Jégou。使用图像 Transformer 走向更深层。arXiv 预印本 arXiv:2103.17239，2021。2 [69] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszko-reit, Llion Jones, Aidan N Gomez， ukasz Kaiser 和 Illia Polosukhin。Attention is all you need。在 NeurIPS，页 5998–6008，2017。1, 2 [70] Alex Wang, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel R Bowman。Glue：用于自然语言理解的多任务基准和分析平台。arXiv 预印本 arXiv:1804.07461，2018。6, 7, 9 [71] Wenhai Wang, Enze Xie, Xiang Li, Deng-Ping Fan, Kaitao Song, Ding Liang, Tong Lu, Ping Luo, and Ling Shao。金字塔视觉 Transformer：一种无需卷积、用于密集预测的通用骨干网络。在 ICCV，2021。2 [72] Han Xu, Zhang Zhengyan, Ding Ning, Gu Yuxian, Liu Xiao, Huo Yuqi, Qiu Jiezhong, Zhang Liang, Han Wentao, Huang Minlie, 等。预训练模型：过去、现在和未来。arXiv 预印本 arXiv:2106.07139，2021。3 [73] Li Yuan, Yunpeng Chen, Tao Wang, Weihao Yu, Yujun Shi, Zi-Hang Jiang, Francis E.H. Tay, Jiashi Feng, and Shuicheng Yan。Tokens-to-token vit：在 imagenet 上从头训练视觉 Transformer。在 ICCV，页 558–567，2021 年 10 月。2 [74] Yanyi Zhang, Xinyu Li, Chunhui Liu, Bing Shuai, Yi Zhu, Biagio Brattoli, Hao Chen, Ivan Marsic, and Joseph Tighe。Vidtr：无卷积的视频 Transformer。在 ICCV，页 13577–13587，2021 年 10 月。2 [75] Ziqi Zhang, Yaya Shi, Chunfeng Yuan, Bing Li, Peijin Wang, Weiming Hu, and Zheng-Jun Zha。用于视频字幕生成的教师推荐学习对象关系图。在 CVPR，页 13278–13288，2020。8 [76] Daquan Zhou, Bingyi Kang, Xiaojie Jin, Linjie Yang, Xi- aochen Lian, Qibin Hou, and Jiashi Feng。Deepvit：迈向更深的视觉 Transformer。arXiv 预印本 arXiv:2103.11886，2021。2 [77] Kaiyang Zhou, Jingkang Yang, Chen Change Loy, and Ziwei Liu。学习为视觉-语言模型进行提示。arXiv 预印本 arXiv:2109.01134，2021。3 [78] Luowei Zhou, Hamid Palangi, Lei Zhang, Houdong Hu, Ja- son Corso, and Jianfeng Gao。用于图像字幕生成和 VQA 的统一视觉-语言预训练。在 AAAI，2020。2, 8 [79] Yukun Zhu, Ryan Kiros, Rich Zemel, Ruslan Salakhutdinov, Raquel Urtasun, Antonio Torralba, and Sanja Fidler。对齐书籍与电影：通过观看电影和阅读书籍实现故事式视觉解释。在 ICCV，页 19–27，2015。6, 14

A. Tokenizer

Given the raw inputs from text, image, and video modalities, modality-specific tokenizers are applied to generate the input token sequences for the Transformer encoder. Here, we use the BPE tokenizer [59] for text modality, the image patch tokenizer [17] for image modality, and the temporal frame patch tokenizer [6] for video modality. These outputted tokens are attached with additional modality type embeddings to identify which modality the raw input belongs to. The tokenizers are illustrated in Fig. 2.

Text Tokenizer. The BPE Tokenizer [59] is employed for text modality. The text inputs are split into sub-words and projected by a linear embedding layer. A learnable 1D positional embedding for text with a max length of 256 is added to the word embeddings. To specify the input modality, an additional trainable textual modality embedding <T> is added to each token embedding. Note that all the text inputs of our model share the same vocabulary.

Image Tokenizer. The image patch tokenizer [17] is utilized as the image tokenizer. The input images are resized to 224×224, and flattened to a sequence of image patches with shapes 16×16, which are further mapped by a linear projection. A sequence of learnable image positional embeddings with a fixed length $14 \times 14 = 196$, and an additional visual modality embedding <V> are also added.

Video Tokenizer. The temporal frame patch tokenizer [6] is used for video tokenization. Each frame of the input video is flattened to image patches with shape 16×16. For a video with N frames ($N = 8$ by default), the number of tokens would be $N \times 14 \times 14$. The spatial positional embeddings for images as well as a 1D temporal position embedding with max length $N = 8$ are added to the video embeddings. Besides, the additional visual modality embedding <V> is added to each token embedding.

B. Implementation Details for Auto-regressive Language Modeling

In the formulation of autoregressive tasks from previous works, each input token attends to all previous tokens, including itself, to predict **the next word**. Unlike previous works, we use <SPE> to predict **the current word**. Fig. 4 shows how we achieve this. In the inference stage, <SPE> is appended to the end of the predicted tokens as the input. The corresponding output is used for prediction. In the training stage, we append several <SPE> tokens after the input sequence. Each <SPE> token is trained to predict a word in the input sequence. The attention mask is designed to make sure that the word tokens do not attend to <SPE> tokens, and each <SPE> token only attends to itself and the previous word tokens. In this way, the training stage and the inference stage are aligned.

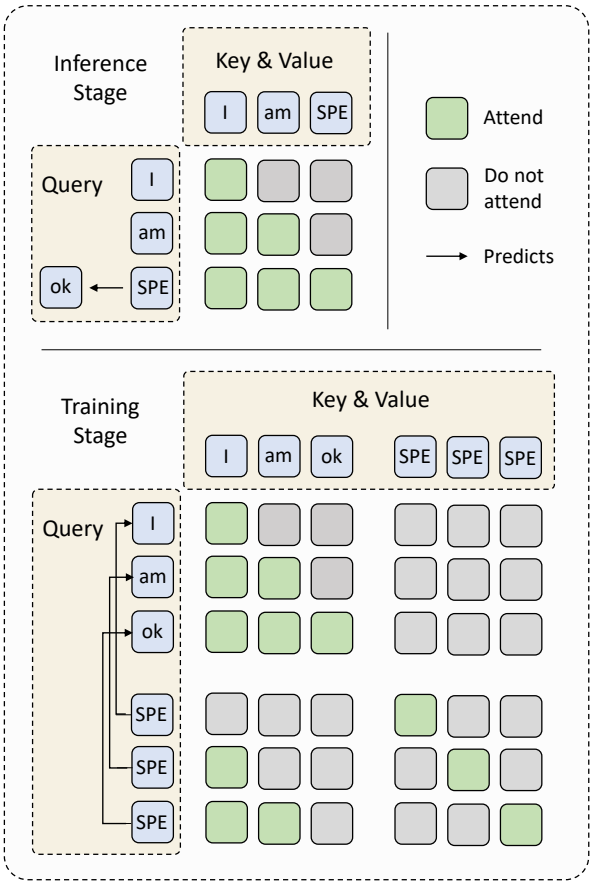


Figure 4. The attention mask of autoregressive language modeling.

C. Prompt Tuning

Four groups of parameters are learnable in prompt-tuning. They are <SPE>, layer normalization parameters, prompt tokens, and linear heads. This section introduces the usage and the number of parameters of each parameter group.

Details Similar to the pre-training stage, <SPE> token is shared for both inputs and targets. Layer normalization weights and biases in each Transformer layer and tokenizer are tuned. Learnable prompts are added to each layer of the Transformer encoder. Specifically, the input prompts of each layer do not come from the output of the previous layer. They are random initialized learnable parameters. For all prompt-tuning experiments, we use 10 learnable prompts for each layer on both inputs. The linear heads only apply for classification tasks. It takes the feature that is to be classified as input, and returns a classification logit. The output probability is a linear combination of the probability from the similarity score and the linear head:

$$P(x, y) \propto \alpha \exp\left(\cos\left(f(x), f(y)\right)/\tau\right) + w^{\top} f(x) + b \quad (4)$$

A. 分词器

给定来自文本、图像和视频模态的原始输入，应用特定于模态的分词器来生成 Transformer 编码器的输入 token 序列。在这里，我们对文本模态使用 BPE 分词器 [59]，对图像模态使用图像 patch 分词器 [17]，对视频模态使用时间帧 patch 分词器 [6]。这些输出的 token 会附加额外的模态类型嵌入，以标识原始输入属于哪种模态。分词器如图 2 所示。

文本分词器。文本模态采用 BPE 分词器 [59]。文本输入被拆分为子词，并通过线性嵌入层进行投影。将一个最大长度为 256 的可学习一维文本位置嵌入添加到词嵌入中。为了指定输入模态，还为每个标记嵌入添加一个额外的可训练文本模态嵌入 <T>。请注意，我们模型的所有文本输入共享同一词汇表。

图像分词器。采用图像块分词器 [17] 作为图像分词器。输入图像被调整为 224×224，并展平成形状为 16×16 的图像块序列，随后通过线性投影进一步映射。还添加了一个固定长度为 $14 \times 14 = 196$ 的可学习图像位置嵌入序列，以及一个额外的视觉模态嵌入 <V>。

视频分词器。时间帧补丁分词器 [6] 用于视频分词。输入视频的每一帧被展平为形状为 16×16 的图像补丁。对于包含 N 帧（默认 $N = 8$）的视频，token 数量为 $N \times 14 \times 14$。将图像的空间位置嵌入以及最大长度为 $N = 8$ 的一维时间位置嵌入添加到视频嵌入中。此外，还将额外的视觉模态嵌入 <V> 添加到每个 token 嵌入中。

B. 自回归语言建模的实现细节

在以往工作的自回归任务表述中，每个输入词元都会关注所有先前词元，包括其自身，以预测下一个词。不同于以往工作，我们使用 <SPE> 来预测当前词。图 4 展示了我们如何实现这一点。在推理阶段，将 <SPE> 追加到已预测词元的末尾作为输入。相应的输出用于预测。在训练阶段，我们在输入序列后追加若干个 <SPE> 词元。每个 <SPE> 词元都被训练用来预测输入序列中的一个词。注意力掩码被设计为确保词词元不会关注 <SPE> 词元，并且每个 <SPE> 词元只关注其自身和先前的词词元。通过这种方式，训练阶段和推理阶段保持一致。

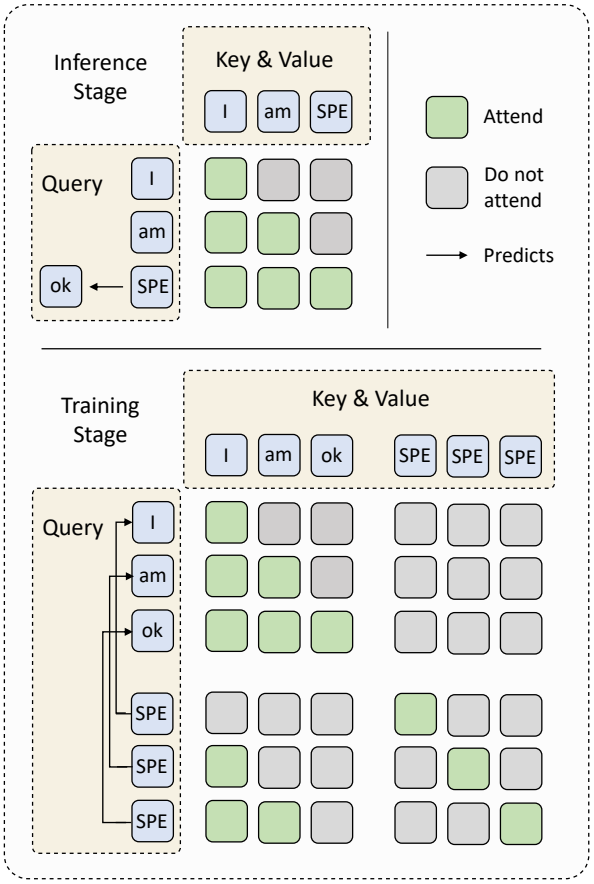


图 4. 自回归语言建模的注意力掩码。

C. 提示调优

在提示调优中，有四组参数是可学习的。它们分别是 <SPE>、层归一化参数、提示词元和线性头。本节介绍每个参数组的用法和参数数量。

细节 与预训练阶段类似，<SPE> 标记在输入和目标中共享。每个 Transformer 层中的层归一化权重和偏置以及分词器都会被调优。可学习提示被添加到 Transformer 编码器的每一层。具体而言，每一层的输入提示并不来自上一层的输出，而是随机初始化的可学习参数。对于所有提示调优实验，我们在两个输入的每一层都使用 10 个可学习提示。线性头仅用于分类任务。它将待分类的特征作为输入，并返回分类 logit。输出概率是来自相似度分数的概率与线性头概率的线性组合：

$$P(x, y) \propto \alpha \exp\left(\cos\left(f(x), f(y)\right)/\tau\right) + w^{\top} f(x) + b \quad (4)$$

	Number of Parameters
<SPE>	768
Layer Norm	41,472
Prompt	184,320
Linear head	768 * num_classes

Table 9. The number of learnable parameters in the prompt-tuning stage. Note that the linear head only applies for classification tasks.

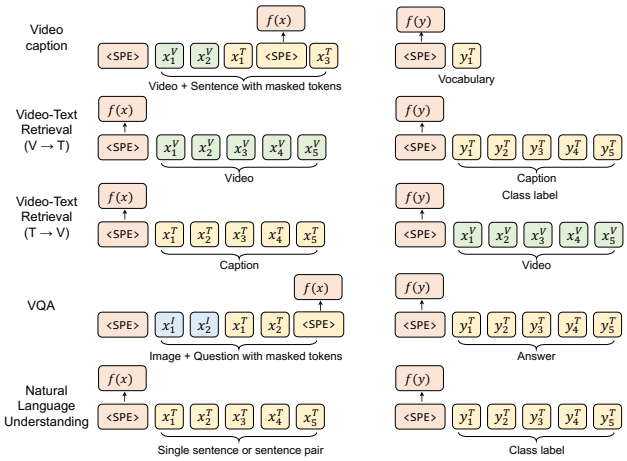


Figure 5. Input and target formats of our novel tasks. For each task, the left column represents the format of input sequence x, and the right column represents the format of the target sequence y. $f(x)$ and $f(y)$ are used to calculate the joint probability distribution. Here, we have omitted the tokenizer and encoder for concision.

where w and b are the weights and bias of the linear, and α is a learnable scalar. w and b are initialized with 0 and α is initialized with 1.

Number of Parameters Tab. 9 shows the number of parameters of each learnable component in prompt-tuning. For tasks other than classification, the number of parameters is 227K in total. When the linear head is added, *e.g.*, in ImageNet-1k classification task, the number is 995K, which is still less than 1% of all the parameters in the pre-training stage.

D. Formulation of Novel Tasks

The generic perceptual modeling makes it easy to convert existing tasks into the unified task formulation of our Uni-Perceiver. Fig. 5 illustrates the input and the output formulations of our novel tasks, which we will describe in details.

Video Caption. Similar to image caption, video caption is modeled as the autoregressive language modeling task with video clues. In this task, the model predicts each word based on the video and its previous words. The input set $\mathcal{X}$ consists of the sequence concatenation of video and the words that have been predicted, followed with a <SPE> token. The output features of the <SPE> token is used to calculate the joint probability with each words in the the vocabulary set $\mathcal{Y}$. Then the word with the highest probability is the predicted word at the current location. Additionally, video caption follows the efficient implementation of autoregressive training introduced in Sec. B.

Video-Text Retrieval. The Video-Text retrieval follows the formulation of Image-Text retrieval task, except that the image sequence is replaced by the video sequence. Specifically, the input sets $\mathcal{X}$ and $\mathcal{Y}$ are composed of video and text sequences respectively. Each sequence in $\mathcal{X}$ and $\mathcal{Y}$ also has a <SPE> token at the beginning. We use the output feature at the <SPE> token as the final representation of the input video or the text to calculate the joint probability distribution.

Visual Question Answering. We formulate the VQA task as a special case of masked language modeling with image clues. The input $x \in \mathcal{X}$ is the combination of image and question sequences. It should be noted that each question ends with a "?" mark and a <SPE> token is appended to the end of the question sequence. The $\mathcal{Y}$ set consists of candidate answer sequences, in which each begins with a <SPE> token too. The joint probability distribution between $\mathcal{X}$ and $\mathcal{Y}$ can be calculated by using the output feature from <SPE> tokens.

Natural Language Understanding. The formulation of natural language understanding task is similar to that of image classification task. $\mathcal{X}$ denotes the set of the input single sentence or the sentence-pair, and $\mathcal{Y}$ is the set contains the textual class label. For example, in SST-2 [87], x denotes the sentence sequence of movie review and $y \in \mathcal{Y} = \{\texttt{great}, \texttt{terrible}\}$ is the sentiment label. In MRPC [81], x instead is the sequence combination of sentence pairs extracted from news, and $y \in \mathcal{Y} = \{\texttt{Yes}, \texttt{No}\}$ is the label to indicate whether the sentence pair are semantically equivalent. We also add <SPE> tokens at the beginning of the sequences x and y, of which output features are used to computed joint probability.

E. Extra pre-training details

Sampling Weight & Batch Size & Loss Tab. 10 lists the batch size and sampling weight of each task and dataset in the pre-training stage. We use cross-entropy loss for language modeling tasks. The other tasks are trained with cross-entropy loss with 0.1 label smoothing. The loss weight of video classification is 0.05, which helps stabilize training in our experiments. The other loss weights are 1.0 by default.

For retrieval tasks like image-text retrieval, we use train-

	Number of Parameters
<SPE>	768
Layer Norm	41,472
Prompt	184,320
Linear head	768 * num_classes

表 9. 提示调优阶段中的可学习参数数量。注意，线性头仅适用于分类任务。

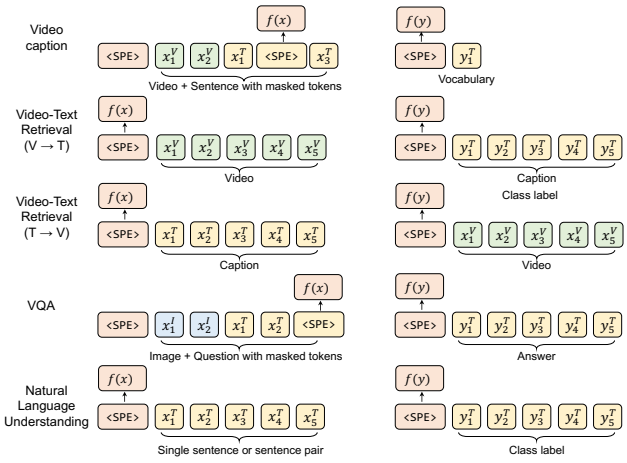


图5. 我们新任务的输入和目标格式。对于每个任务，左列表示输入序列 x 的格式，右列表示目标序列 y 的格式。$f(x)$ 和 $f(y)$ 用于计算联合概率分布。为简洁起见，这里省略了分词器和编码器。

其中 w 和 b 是线性层的权重和偏置，α 是一个可学习标量。w 和 b 初始化为 0，α 初始化为 1。

参数数量。表9展示了提示调优中每个可学习组件的参数数量。对于分类以外的任务，参数总数为227K。当添加线性头时，例如在 ImageNet-1k 分类任务中，参数数量为995K，仍少于预训练阶段所有参数的1%。

D. 新任务的表述

通用感知建模使得将现有任务转换为我们 Uni-Perceiver 的统一任务形式变得容易。图 5 展示了我们新任务的输入和输出形式，下面将对其进行详细描述。

视频字幕。与图像字幕类似，视频字幕被建模为带有视频线索的自回归语言建模任务。在该任务中，模型基于视频及其先前的词来预测每个词。输入集 $\mathcal{X}$ 由视频和已预测单词的序列拼接组成，随后接一个 <SPE> 标记。<SPE> 标记的输出特征用于计算其与词汇表 $\mathcal{Y}$ 中每个单词的联合概率。然后，概率最高的单词即为当前位置的预测单词。此外，视频字幕生成遵循第 B 节中介绍的自回归训练的高效实现。

视频-文本检索。视频-文本检索遵循图像-文本检索任务的公式化方法，只是将图像序列替换为视频序列。具体而言，输入集合 $\mathcal{X}$ 和 $\mathcal{Y}$ 分别由视频序列和文本序列组成。$\mathcal{X}$ 和 $\mathcal{Y}$ 中的每个序列开头也都有一个 <SPE> 标记。我们使用 <SPE> 标记处的输出特征作为输入视频或文本的最终表示，以计算联合概率分布。

视觉问答。我们将 VQA 任务表述为一种带有图像线索的掩码语言建模特例。输入 $x \in \mathcal{X}$ 是图像序列和问题序列的组合。需要注意的是，每个问题都以“?”标记结尾，并且在问题序列末尾追加一个 <SPE> 标记。$\mathcal{Y}$ 集合由候选答案序列组成，其中每个序列也以一个 <SPE> 标记开头。$\mathcal{X}$ 和 $\mathcal{Y}$ 之间的联合概率分布可以通过使用来自 <SPE> 标记的输出特征来计算。

自然语言理解。自然语言理解任务的形式化表述与图像分类任务类似。$\mathcal{X}$ 表示输入的单句或句子对的集合，$\mathcal{Y}$ 是包含文本类别标签的集合。例如，在 SST-2 [87] 中，x 表示电影评论的句子序列，$y \in \mathcal{Y} = \{\text{好}, \text{糟糕}\}$ 是情感标签。在 MRPC [81] 中，x 则是从新闻中提取的句子对的序列组合，$y \in \mathcal{Y} = \{\text{是}, \text{否}\}$ 是用于指示句子对在语义上是否等价的标签。我们还在序列 x 和 y 的开头添加 <SPE> 标记，其输出特征用于计算联合概率。

E. 额外的预训练细节

采样权重、批大小与损失 表10列出了预训练阶段每个任务和数据集的批大小及采样权重。我们对语言建模任务使用交叉熵损失。其他任务使用带有0.1标签平滑的交叉熵损失进行训练。视频分类的损失权重为0.05，这在我们的实验中有助于稳定训练。其他损失权重默认均为1.0。

对于图像-文本检索等检索任务，我们使用 train-

Task	Dataset	Batch Size	Sampling Weight
Image Classification	ImageNet-21k [15]	64	0.333
Video Classification	Kinetics-700 [27]	4	0.0925
	Moments in Time [49]	24	0.0185
Auto-encoding LM	Books&Wiki [79]	64	0.07775
	YFCC [26]	64	0.02778
	CC12M [8]	64	0.02778
	CC3M [60]	64	0.01389
	Visual Genome [30]	64	0.01389
	COCO Caption [11]	64	0.01389
	SBU [50]	64	0.01389
	PAQ [35]	512	0.0222
Auto-regressive LM	Books&Wiki [79]	64	0.07775
	YFCC [26]	56	0.02778
	CC12M [8]	56	0.02778
	CC3M [60]	56	0.01389
	Visual Genome [30]	56	0.01389
	COCO Caption [11]	56	0.01389
	SBU [50]	56	0.01389
	PAQ [35]	400	0.0222
Retrieval	YFCC [26]	128	0.02778
	CC12M [8]	128	0.02778
	CC3M [60]	128	0.01389
	Visual Genome [30]	128	0.01389
	COCO Caption [11]	128	0.01389
	SBU [50]	128	0.01389
	PAQ [35]	512	0.0222

Table 10. Ingredients and hyper-parameters for our pre-training.

ing samples in the same batch as negative samples, whose typical size is 127 except the PAQ dataset. Note that we do not use memory bank and do not gather feature across GPU devices to provide more negative samples, which may further promote the performance of retrieval tasks.

Data Augmentation We apply augmentation techniques to image and video modalities to avoid overfitting. For images in ImageNet-21k dataset, we apply augmentation same as [67]. Rand-Aug[80], random erasing [91], mixup [90] and cutmix [89] are used simultaneously. For images in other datasets, we resize the images to the short edge size of 256, and then a 224×224 region is cropped randomly from the resized images during training. During inference, the random crop is replaced with center crop operation. For all Video inputs, we apply the same augmentations used in [6]. We use clips of size $8 \times 224 \times 224$ for Kinetics-700 and Kinetics-400, and $3 \times 224 \times 224$ for Moments in Time. The temporal sample rate is 32. We use a single temporal clip. During training, the start frame is randomly picked if the video is longer than the clip. In the training stage, we first resize the shorter side of the video to a random value in $[256, 320]$, then we randomly sample a 224×224 crop. In the test stage, the short side of the video is resized to 224. For classification tasks, We use 3 spatial crops with size 224×224 to cover a larger range of content and average their logits for evaluation. For video captioning task, we use center crop after resizing.

Data Parallel for Vocabulary and Class Labels Naive implementation of language modeling and ImageNet-21k classification is impractical due to memory limitation. In language modeling, we need to compare the feature of a <SPE> token in a sequence to the feature of each token in our tokenizer. Since the vocabulary size is large, we apply data parallel for the vocabulary set. A similar method is used for class labels of ImageNet-21k.

Removing Overlap For the training set of K700 participating in pre-training, we remove those videos overlapping with validation set of K400.

F. Licences of Datasets

ImageNet-21K [15] is subject to the ImageNet terms of use [88].

Kinetics-700 [86] & **Kinetics-400** [27] The kinetics dataset is licensed by Google Inc. under a Creative Commons Attribution 4.0 International License.

BooksCorpus [79] Replicate Toronto BookCorpus is open-source and licensed under GNU GPL, Version 3.

Wikipedia Most of Wikipedia's text is co-licensed under the Creative Commons Attribution-ShareAlike 3.0 Unported License (CC BY-SA) and the GNU Free Documentation License (GFDL) (unversioned, with no invariant sections, front-cover texts, or back-cover texts). Some text has been imported only under CC BY-SA and CC BY-SA-compatible license and cannot be reused under GFDL.

YFCC [26] All the photos and videos provided in YFCC dataset are licensed under one of the Creative Commons copyright licenses.

CC12M [8] is licensed under the Terms of Use of Conceptual 12M [84].

CC3M [60] is licensed under the Conceptual Captions Terms of Use [85].

Visual Genome [30] is licensed under a Creative Commons Attribution 4.0 International License [83].

COCO Caption [11] The images are subject to the Flickr terms of use [82].

SBU Caption [50] The images are subject to the Flickr terms of use [82].

PAQ [35] is licensed under the Attribution-NonCommercial 4.0 International License.

Appendix References

[80] Ekin D Cubuk, Barret Zoph, Jonathon Shlens, and Quoc V Le. Randaugment: Practical automated data augmentation with a reduced search space. In *CVPR Workshops*, pages 702–703, 2020.

Task	Dataset	Batch Size	Sampling Weight
Image Classification	ImageNet-21k [15]	64	0.333
Video Classification	Kinetics-700 [27]	4	0.0925
	Moments in Time [49]	24	0.0185
Auto-encoding LM	Books&Wiki [79]	64	0.07775
	YFCC [26]	64	0.02778
	CC12M [8]	64	0.02778
	CC3M [60]	64	0.01389
	Visual Genome [30]	64	0.01389
	COCO Caption [11]	64	0.01389
	SBU [50]	64	0.01389
	PAQ [35]	512	0.0222
Auto-regressive LM	Books&Wiki [79]	64	0.07775
	YFCC [26]	56	0.02778
	CC12M [8]	56	0.02778
	CC3M [60]	56	0.01389
	Visual Genome [30]	56	0.01389
	COCO Caption [11]	56	0.01389
	SBU [50]	56	0.01389
	PAQ [35]	400	0.0222
Retrieval	YFCC [26]	128	0.02778
	CC12M [8]	128	0.02778
	CC3M [60]	128	0.01389
	Visual Genome [30]	128	0.01389
	COCO Caption [11]	128	0.01389
	SBU [50]	128	0.01389
	PAQ [35]	512	0.0222

表 10. 我们预训练的组成要素和超参数。

将同一批次中的样本作为负样本，其典型大小为 127，PAQ 数据集除外。请注意，我们不使用内存库，也不跨 GPU 设备聚合特征以提供更多负样本，而这些做法可能会进一步提升检索任务的性能。

数据增强 我们对图像和视频模态应用增强技术以避免过拟合。对于 ImageNet-21k 数据集中的图像，我们采用与 [67] 相同的增强方法。同时使用 Rand-Aug[80]、随机擦除 [91]、mixup [90] 和 cutmix [89]。对于其他数据集中的图像，我们将图像调整为短边大小为 256，然后在训练期间从调整后的图像中随机裁剪一个 224×224 的区域。在推理期间，随机裁剪被替换为中心裁剪操作。对于所有视频输入，我们采用 [6] 中使用的相同增强方法。对于 Kinetics-700 和 Kinetics-400，我们使用大小为 $8 \times 224 \times 224$ 的片段；对于 Moments in Time，我们使用 $3 \times 224 \times 224$。时间采样率为 32。我们使用单个时间片段。在训练期间，如果视频长度超过片段长度，则随机选择起始帧。在训练阶段，我们首先将视频的较短边调整为 [256, 320] 中的随机值，然后随机采样一个 224×224 的裁剪区域。在测试阶段，将视频的短边调整为 224。对于分类任务，我们使用 3 个大小为 224×224 的空间裁剪，以覆盖更大范围的内容，并对它们的 logits 取平均用于评估。对于视频字幕生成任务，我们在调整大小后使用中心裁剪。

用于词汇表和类别标签的数据并行：语言建模和 ImageNet-21k 的朴素实现

由于内存限制，分类并不实际。在语言建模中，我们需要将序列中某个 <SPE> 标记的特征与分词器中每个标记的特征进行比较。由于词汇表规模很大，我们对词汇表集合应用数据并行。类似的方法也用于 ImageNet-21k 的类别标签。

去除重叠 对于参与预训练的 K700 训练集，我们移除了那些与 K400 验证集重叠的视频。

F. 数据集许可证

ImageNet-21K [15] 受 ImageNet 使用条款 [88] 约束。

Kinetics-700 [86] & Kinetics-400 [27] Kinetics 数据集由 Google Inc. 根据知识共享署名 4.0 国际许可证授权。

BooksCorpus [79] 复制版 Toronto BookCorpus 是开源的，并根据 GNU GPL 第 3 版授权。

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CC12M [8] 根据 Conceptual 12M [84] 的使用条款获得许可。

CC3M [60] 根据 Conceptual Captions 使用条款 [85] 获得许可。

Visual Genome [30] 采用知识共享署名 4.0 国际许可协议 [83] 授权。

COCO Caption [11] 这些图像受 Flickr 使用条款 [82] 的约束。

SBU Caption [50] 图像受 Flickr 使用条款 [82] 约束。

PAQ [35] 根据署名-非商业性使用 4.0 国际许可协议授权。

附录参考文献

[80] Ekin D Cubuk、Barret Zoph、Jonathon Shlens 和 Quoc V Le。Randaugment：具有缩小搜索空间的实用自动化数据增强。载于 CVPR Workshops，第 702–703 页，2020 年。

[81] William B Dolan and Chris Brockett. Automatically constructing a corpus of sentential paraphrases. In *Proceedings of the Third International Workshop on Paraphrasing (IWP2005)*, 2005.

[82] Inc. Flickr. Flickr terms & conditions of use. https://www.flickr.com/help/terms.

[83] Ranjay Krishna. Visual genome terms & conditions of use. https://visualgenome.org/about.

[84] Google LLC. Conceptual 12m terms & conditions of use. https://github.com/google-research-datasets/conceptual-12m/blob/main/LICENSE.

[85] Google LLC. Conceptual captions terms & conditions of use. https://github.com/google-research-datasets/conceptual-captions/blob/master/LICENSE.

[86] Lucas Smaira, João Carreira, Eric Noland, Ellen Clancy, Amy Wu, and Andrew Zisserman. A short note on the kinetics-700-2020 human action dataset. *arXiv preprint arXiv:2010.10864*, 2020.

[87] Richard Socher, Alex Perelygin, Jean Wu, Jason Chuang, Christopher D Manning, Andrew Y Ng, and Christopher Potts. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of the 2013 conference on empirical methods in natural language processing*, pages 1631–1642, 2013.

[88] Princeton University and Stanford University. Imagenet terms & conditions of use. https://image-net.org/download.

[89] Sangdoo Yun, Dongyoon Han, Seong Joon Oh, Sanghyuk Chun, Junsuk Choe, and Youngjoon Yoo. Cutmix: Regularization strategy to train strong classifiers with localizable features. In *ICCV*, pages 6023–6032, 2019.

[90] Hongyi Zhang, Moustapha Cisse, Yann N Dauphin, and David Lopez-Paz. mixup: Beyond empirical risk minimization. *arXiv preprint arXiv:1710.09412*, 2017.

[91] Zhun Zhong, Liang Zheng, Guoliang Kang, Shaozi Li, and Yi Yang. Random erasing data augmentation. In *AAAI*, volume 34, pages 13001–13008, 2020.

[81] William B Dolan 和 Chris Brockett。自动构建句子级释义语料库。载于第三届国际释义研讨会（IWP2005）论文集，2005。 [82] Inc. Flickr。Flickr 使用条款与条件。https:// www.flickr.com/help/terms。 [83] Ranjay Krishna。Visual Genome 使用条款与条件。https://visualgenome.org/about。 [84] Google LLC。Conceptual 12m 使用条款与条件。https : / / github . com / google - research - datasets / conceptual - 12m / blob / main / LICENSE。 [85] Google LLC。Conceptual Captions 使用条款与条件。https : / / github . com / google - research - datasets / conceptual - captions / blob/master/LICENSE。 [86] Lucas Smaira、João Carreira、Eric Noland、Ellen Clancy、Amy Wu 和 Andrew Zisserman。关于 kinetics-700-2020 人类动作数据集的简短说明。arXiv 预印本 arXiv:2010.10864，2020。 [87] Richard Socher、Alex Perelygin、Jean Wu、Jason Chuang、Christopher D Manning、Andrew Y Ng 和 Christopher Potts。用于情感树库上语义组合性的递归深度模型。载于 2013 年自然语言处理经验方法会议论文集，第 1631–1642 页，2013。 [88] Princeton University 和 Stanford University。Imagenet 使用条款与条件。https://image-net.org/ download。 [89] Sangdoo Yun、Dongyoon Han、Seong Joon Oh、Sanghyuk Chun、Junsuk Choe 和 Youngjoon Yoo。Cutmix：用于训练具有可定位特征的强分类器的正则化策略。载于 ICCV，第 6023–6032 页，2019。 [90] Hongyi Zhang、Moustapha Cisse、Yann N Dauphin 和 David Lopez-Paz。mixup：超越经验风险最小化。arXiv 预印本 arXiv:1710.09412，2017。 [91] Zhun Zhong、Liang Zheng、Guoliang Kang、Shaozi Li 和 Yi Yang。随机擦除数据增强。载于 AAAI，第 34 卷，第 13001–13008 页，2020。